



វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា

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IN
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**MULTI-TASK DEEP LEARNING FOR FOOD
RECOGNITION AND NUTRITION ESTIMATION
FROM ASIAN FOOD IMAGES**

A THESIS SUBMITTED BY

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UNDER CO-SUPERVISION OF

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វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា**



**និក្ខេបបទបរិញ្ញាបត្រជាន់ខ្ពស់
របស់និស្សិត ត្រីវិទ្យា ម៉េងរាល**

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ក្នុងរយៈពេលប៉ុន្មានឆ្នាំចុងក្រោយនេះ ការតាមដានការទទួលទានអាហារូបត្ថម្ភបានក្លាយជាបញ្ហា កាន់តែសំខាន់ ដើម្បីលើកកម្ពស់ទម្លាប់នៃការទទួលទានអាហារដែលមានសុខភាពល្អ និងកាត់បន្ថយហានិភ័យនៃជំងឺដែលបណ្តាលមកពីរបបអាហារ។ ការរក្សារបៀបរស់នៅប្រកបដោយសុខភាពល្អ តម្រូវឱ្យមានការទទួលទានអាហារដែលមានគុណភាពផ្នែកអាហារូបត្ថម្ភ។ ការស្រាវជ្រាវនេះបង្ហាញពី Multi-Task Deep Learning បែប End-to-End សម្រាប់ការចាត់ថ្នាក់អាហារ និងការប៉ាន់ប្រមាណតម្លៃអាហារូបត្ថម្ភក្នុងពេលតែមួយ ដោយប្រើរូបភាពអាហារអាស៊ីពីរវិមាត្រ (2D)។

ការសិក្សានេះបានប្រើសំណុំទិន្នន័យ NutritionVerse-Real ដែលត្រូវបានសម្អាត និងបង្កើនទិន្នន័យបន្ថែមដោយប្រើវិធីសាស្ត្រ Data Augmentation ដោយផ្ដោតលើប្រភេទអាហារទូទៅចំនួន ៤២ ប្រភេទ បន្ទាប់ពីដកចេញនូវប្រភេទដែលមានទិន្នន័យមិនពេញលេញ។ ស្ថាបត្យកម្មដែលបានស្នើប្រើបណ្តាញសរសៃប្រសាទបែបកុងវ៉ូលូសិន (Convolutional Neural Network) រួមមួយ (Shared Backbone) ជាមួយសាខាព្យាករណ៍ពីរ។ សាខាទីមួយជាគំរូពហុចាត់ថ្នាក់ (Multi-label Classification) ដែលប្រើមុខងារ Sigmoid Activation ដើម្បីកំណត់អត្តសញ្ញាណសមាសធាតុអាហារដែលមាននៅក្នុងមុខម្ហូប។ សាខាទីពីរជាគំរូតំរង់ (Regression) សម្រាប់ប៉ាន់ប្រមាណអាហារូបត្ថម្ភនៅកម្រិតមុខម្ហូបទាំងមូល។ គំរូត្រូវបានបណ្តុះបណ្តាលដោយប្រើមុខងារទម្ងន់ខាតបង់រួម (Weighted Combined Loss Function) ដោយបញ្ចូល Weighted Binary Cross-Entropy Loss សម្រាប់ការចាត់ថ្នាក់អាហារ និង Huber Loss សម្រាប់ការប៉ាន់ប្រមាណអាហារូបត្ថម្ភ ដើម្បីធ្វើឱ្យមានគុណភាពរវាងភារកិច្ចទាំងពីរអោយបានប្រសើរ។ ការសិក្សានេះបានវាយតម្លៃដោយប្រើ Convolutional Neural Network backbone ចំនួនពីរ គឺ MobileNetV2 និង ResNet50 ដែលជាគំរូស្រង់លក្ខណៈពិសេស (Feature Extractors) នៅក្នុងការងារពហុភារកិច្ចនេះ។

លទ្ធផលពិសោធន៍បង្ហាញថា ResNet50 ទទួលបានលទ្ធផលប្រសើរជាងសម្រាប់ការសម្គាល់ប្រភេទអាហារជារួម ដោយសម្រេចបាន Micro Recall ស្មើ 0.5273, Micro F1-score ស្មើ 0.4979 និង Macro F1-score ស្មើ 0.3905 ខណៈដែល MobileNetV2 ទទួលបាន Micro Precision និង Hamming Accuracy ខ្ពស់ជាងបន្តិច។ សម្រាប់ការប៉ាន់ប្រមាណតម្លៃអាហារូបត្ថម្ភ គំរូទាំងពីរទទួលបានតម្លៃ R^2 វិជ្ជមានសម្រាប់គោលដៅអាហារូបត្ថម្ភទាំងប្រាំ។ MobileNetV2 ទទួលបានតម្លៃ R^2 ស្មើ 0.173 សម្រាប់កាឡូរី, 0.190 សម្រាប់ខ្លាញ់, 0.279 សម្រាប់កាបូអ៊ីដ្រាត, 0.054 សម្រាប់ប្រូតេអ៊ីន និង 0.613 សម្រាប់សូដ្យូម។ ចំណែក ResNet50 ទទួលបានតម្លៃ R^2 ស្មើ 0.166 សម្រាប់កាឡូរី, 0.049 សម្រាប់ខ្លាញ់, 0.291 សម្រាប់កាបូអ៊ីដ្រាត, 0.072 សម្រាប់ប្រូតេអ៊ីន និង 0.691 សម្រាប់សូដ្យូម។ ការប្រៀបធៀបបង្ហាញថា MobileNetV2 ផ្តល់លទ្ធផលល្អជាងសម្រាប់ការប៉ាន់ប្រមាណកាឡូរី និងខ្លាញ់ ខណៈដែល ResNet50 ផ្តល់លទ្ធផលប្រសើរជាងសម្រាប់កាបូអ៊ីដ្រាត ប្រូតេអ៊ីន និងសូដ្យូម។ លទ្ធផលទាំងនេះបញ្ជាក់ថា ការកំណត់ទម្ងន់ខាតបង់តាមស្ថាបត្យកម្មនីមួយៗអាចធ្វើឱ្យប្រសើរឡើងនូវគុណភាពរវាងការសម្គាល់ប្រភេទ

អាហារ និងការប៉ាន់ប្រមាណតម្លៃអាហារូបត្ថម្ភ។ ទោះជាយ៉ាងណាការប៉ាន់ប្រមាណសារធាតុចិញ្ចឹមពីរូបភាពអាហារពីរីមាត្រតែមួយនៅតែមានការលំបាក ជាពិសេសសម្រាប់សារធាតុចិញ្ចឹមដែលអាស្រ័យខ្លាំងលើទំហំចំណែកអាហារ និងបរិមាណអាហារ។ លទ្ធផលនៃការសិក្សានេះបង្ហាញពីសក្តានុពលនៃគំរូ Multi-Task Deep Learning សម្រាប់ការសម្គាល់អាហារ និងការប៉ាន់ប្រមាណតម្លៃអាហារូបត្ថម្ភក្នុងពេលតែមួយ ហើយផ្តល់ជាមូលដ្ឋានសម្រាប់ការអភិវឌ្ឍប្រព័ន្ធតាមដានរបបអាហារដោយស្វ័យប្រវត្តិ និងការស្រាវជ្រាវបន្ថែមលើការវាយតម្លៃអាហារូបត្ថម្ភដោយផ្អែកលើរូបភាព។

ABSTRACT

In recent years, monitoring nutritional intake has become increasingly important for promoting healthy dietary habits and reducing the risk of diet-related diseases. Maintaining a healthy lifestyle requires the consumption of nutritionally balanced foods. This research presents a deep learning-based end-to-end multi-task learning framework for simultaneous food recognition and nutrition estimation from 2D food images of Asian food dishes.

A cleaned and augmented subset of the NutritionVerse-Real dataset was used, focusing on 42 common food types after removing incomplete classes. The proposed architecture employed a shared convolutional neural network backbone with two prediction branches: a multi-label classification head using sigmoid activation to identify food components inside a dish and a regression head to estimate nutritional values at the dish level. The model was trained using a weighted combined loss function that combines weighted binary cross-entropy for food classification and Huber loss for nutrition regression to balance the two learning tasks effectively. Two convolutional neural network backbones, MobileNetV2 and ResNet50 were evaluated as feature extractors within this multi-task framework.

The experimental results showed that ResNet50 achieved stronger overall food recognition performance, obtaining a micro recall of 0.5273, a micro F1-score of 0.4979, and a macro F1-score of 0.3905, while MobileNetV2 achieved slightly higher micro precision and Hamming accuracy. For nutrition estimation, both architectures produced positive R^2 values across all five nutritional targets. MobileNetV2 achieved R^2 values of 0.173 for calories, 0.190 for fats, 0.279 for carbohydrates, 0.054 for protein, and 0.613 for sodium. ResNet50 achieved corresponding R^2 values of 0.166, 0.049, 0.291, 0.072, and 0.691. ResNet50 therefore provided stronger prediction performance for carbohydrates, protein, and sodium, whereas MobileNetV2 remained more accurate for calories and fats. These findings demonstrate that task-specific loss balancing can substantially improve regression performance and that the optimal classification-regression trade-off may differ according to backbone architecture. The results also confirm that nutrition estimation from single 2D food images remains challenging, particularly for nutrients strongly influenced by portion size and food volume. These findings provide a foundation for future automated dietary monitoring systems and further research on image-based nutrition assessment.

ABBREVIATIONS AND SYMBOLS

AI:	Artificial Intelligence
ANN:	Artificial Neural Network
AUC:	Area Under the Receiver Operating Characteristic Curve
BCE:	Binary Cross-Entropy
CNN:	Convolutional Neural Network
DL:	Deep Learning
F1:	F1-score
GPU:	Graphics Processing Unit
ImageNet:	Image Database for Visual Object Recognition
MAE:	Mean Absolute Error
ML:	Machine Learning
MSE:	Mean Squared Error
MTL:	Multi-Task Learning
RGB:	Red, Green, Blue
RMSE:	Root Mean Square Error
ReLU:	Rectified Linear Unit
ROC:	Receiver Operating Characteristic
R²:	Coefficient of Determination
SDGs:	Sustainable Development Goals
WBCE:	Weighted Binary Cross-Entropy
n:	Number of samples (observations)
N:	Total number of samples in the dataset
L:	Number of food classes (labels)
x:	Input food image
F:	Feature representation extracted by the shared backbone
y:	Ground-truth value
$\hat{y}$:	Predicted value
y_i:	Ground-truth value of the i^{th} sample
$\hat{y}_i$:	Predicted value of the i^{th} sample
$\bar{y}$:	Mean of the ground-truth values

TP:	True Positive
TN:	True Negative
FP:	False Positive
FN:	False Negative
L_{WBCE}:	Weighted Binary Cross-Entropy loss
L_{Huber}:	Huber loss
L_{Total}:	Combined multi-task loss
w_1:	Weight assigned to the classification loss
w_2:	Weight assigned to the regression loss
θ:	Huber loss threshold parameter
δ:	Sigmoid activation function
w:	Learnable weight matrix
b:	Bias vector
Z:	Concatenated feature vector for classification-conditioned regression

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CHAPTER 1

INTRODUCTION

1.1. Research Background

Dietary behavior is widely recognized as one of the most important modifiable factors influencing human health and well-being. Maintaining a nutritionally balanced diet plays a crucial role in supporting normal physiological functions, preventing chronic diseases, and improving overall quality of life [1]. Conversely, unhealthy eating habits, including excessive consumption of energy-dense and highly processed foods, have contributed to a rapid increase in diet-related health problems worldwide. According to the World Health Organization (WHO), non-communicable diseases (NCDs), including obesity, type 2 diabetes, cardiovascular diseases, and hypertension, account for approximately 74% of global deaths, making them the leading cause of mortality worldwide [2]. Consequently, improving dietary monitoring and promoting healthier eating behaviors have become major public health priorities across Asia and worldwide.

The burden of diet-related diseases has become particularly significant in Southeast Asia [3], where rapid urbanization, economic development, and changing lifestyles have substantially altered dietary patterns. Traditional diets consisting primarily of fresh and minimally processed foods are increasingly being replaced by meals containing excessive calories, sodium, saturated fat, and added sugars. These dietary changes have contributed to the growing prevalence of obesity, diabetes, hypertension, and cardiovascular diseases throughout the region [1]. Similar trends have also been observed in Cambodia, where increasing urbanization and changes in food consumption behaviors have created new challenges for nutritional management and chronic disease prevention. These observations highlight the growing need for accessible technologies capable of supporting daily dietary monitoring and promoting healthier nutritional choices. Accurate assessment of food intake is fundamental for nutrition management, disease prevention, and personalized healthcare. Traditionally, dietary assessment has relied on methods such as food diaries, 24-hour dietary recalls, and food frequency questionnaires. Although these approaches are widely adopted in nutritional research and clinical practice, they depend heavily on users' memory, manual recording, and subjective estimation of food portions. Such methods are often time-consuming, labor-intensive, and susceptible to reporting bias, leading to inaccurate estimation of dietary

intake [4]. Furthermore, estimating the nutritional composition of mixed dishes commonly consumed in Asian countries presents an additional challenge because a single meal frequently contains multiple food components with varying ingredient quantities and preparation methods.

Recent advances in artificial intelligence, particularly deep learning and computer vision, have created new opportunities for automated dietary assessment. Instead of requiring users to manually record food consumption, image-based approaches enable nutritional information to be estimated directly from food photographs captured using smartphones or other imaging devices. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in various computer vision applications, including image classification, object detection, and semantic segmentation, making them well suited for food image analysis [5]. Transfer learning has further improved the practicality of these approaches by enabling pretrained deep learning models to learn effective visual representations from relatively limited food image datasets [6].

Existing research has demonstrated promising performance for food recognition, portion estimation, calorie estimation, and food weight prediction using deep learning techniques. However, most previous studies focus on only a single task, such as food classification or calorie estimation, while relatively few investigate the simultaneous prediction of food categories and multiple nutritional attributes within a unified framework. In addition, many existing datasets primarily contain Western food images or relatively simple food scenes, limiting their applicability to the complex mixed dishes commonly found in Asian cuisine. Furthermore, although several studies have explored multi-task learning, limited research has systematically compared lightweight and deep convolutional backbone architectures under identical experimental settings for simultaneous food recognition and nutrition estimation [4].

To address these limitations, this research proposes a deep learning-based end-to-end multi-task learning framework for simultaneous food recognition and nutrition estimation from two-dimensional RGB images of Asian food. The proposed framework employs a shared convolutional neural network backbone with two task-specific prediction branches. The classification branch performs multi-label food recognition to identify multiple food components present within a dish, while the regression branch estimates dish-level nutritional values, including calories, fats, carbohydrates, protein, and sodium. Two widely adopted transfer learning architectures, MobileNetV2 and ResNet50, are investigated and compared under identical experimental settings to evaluate the trade-off between classification performance, nutrition estimation accuracy, and computational efficiency. By integrating food

recognition and nutrition estimation into a unified learning framework, this research aims to contribute to the development of practical and intelligent dietary assessment systems capable of supporting healthier lifestyles and future mobile health applications.

1.2. Problem Statements

Despite the growing importance of dietary monitoring for disease prevention and health management, accurately estimating nutritional intake from daily meals remains a challenging task. Traditional dietary assessment methods, including food diaries, 24-hour dietary recalls, and food frequency questionnaires, rely heavily on manual recording and users' ability to accurately remember and estimate food consumption. These approaches are not only time-consuming but are also susceptible to reporting errors, recall bias, and inaccurate portion estimation, reducing the reliability of nutritional assessment, particularly in long-term dietary monitoring [4]. Recent advances in computer vision and deep learning have provided promising alternatives for automated dietary assessment through food image analysis. Numerous studies have demonstrated encouraging performance in food recognition, calorie estimation, food weight estimation, and portion size estimation using convolutional neural networks. Nevertheless, most existing approaches address these tasks independently, focusing primarily on a single prediction objective rather than simultaneously recognizing food components and estimating multiple nutritional values. Such task-specific approaches fail to fully exploit the strong relationship between food identity and nutritional composition, potentially limiting the overall effectiveness of automated dietary assessment systems [4] [7]. Furthermore, nutrition estimation remains substantially more challenging than conventional food classification. While food recognition aims to identify the food categories present in an image, accurate nutrition estimation requires quantitative information regarding ingredient composition, serving size, food volume, and preparation methods. These factors are often difficult to infer from a single two-dimensional RGB image, particularly for mixed dishes containing multiple food components. Consequently, many existing methods exhibit limited regression performance when predicting nutritional values that are strongly influenced by portion size, such as calories, fats, and protein [8] [9].

Another limitation of current research is the lack of comprehensive evaluation using Asian food datasets. Compared with Western cuisine, Asian meals frequently consist of multiple ingredients served together within a single dish, creating a more complex multi-label recognition problem. In addition, relatively few studies have systematically investigated the

influence of different convolutional neural network backbone architectures on simultaneous food recognition and nutrition estimation under identical experimental settings. As a result, it remains unclear whether lightweight architectures designed for computational efficiency can achieve comparable or even superior performance to deeper and more computationally intensive networks in multi-task learning scenarios. To address these research gaps, this study proposes a deep learning-based end-to-end multi-task learning framework for simultaneous food recognition and nutrition estimation using two-dimensional RGB images of Asian food. The proposed framework employs a shared convolutional neural network backbone with separate classification and regression branches, where food recognition outputs are incorporated into the nutrition estimation process to exploit the relationship between food identity and nutritional composition. In addition, this study systematically compares MobileNetV2 and ResNet50 using identical datasets, preprocessing procedures, training strategies, and evaluation metrics to investigate the trade-off between classification performance, nutrition estimation accuracy, and computational efficiency. It is expected that the findings of this research will contribute to the development of more effective and practical image-based dietary assessment systems.

1.3. Research Objectives

To address these gaps, this study proposes an end-to-end deep learning framework designed to estimate per-dish nutritional values from a single 2D food image. The proposed solution involves:

- Develop an end-to-end multi-task deep learning framework that jointly performs multi-label food component recognition and dish-level nutritional value estimation from a single 2D food image, using a shared CNN backbone with two parallel task-specific branches.
- Implement a classification-conditioned regression architecture in which the nutritional estimation branch receives food composition predictions classification hidden activations and output probabilities as explicit conditioning inputs, encoding the causal relationship between food identity and nutritional content.
- Design and validate a per-nutrient weighted Huber regression loss that assigns differentiated gradient weights to the five nutritional targets based on their empirical predictability from 2D visual features, improving training signal quality for harder-to-predict macronutrients.

- Implement a hybrid class imbalance mitigation strategy combining targeted data augmentation for minority food classes and class-weighted Binary Cross-Entropy loss, and evaluate their combined effect on food recognition performance across the class distribution.
- Systematically compare MobileNetV2 and ResNet50 as backbone architectures within the identical multi-task framework to characterize the accuracy efficiency trade-off and the impact of model capacity on multi-task performance with limited training data.

1.4. Scope and Limitation

1.4.1. Scope of the Research

The scope of this research is strictly defined by the following parameters to ensure a focused and rigorous analysis:

- **Dataset:** The study utilizes a refined subset of the NutritionVerse-Real dataset, focusing exclusively on 42 common Asian food categories [10]. This includes dishes with complex textures and mixed ingredients typical of regional diets.
- **Task:** Two tasks are jointly addressed: (1) multi-label food component recognition predicting the presence/absence of each of 42 food types per dish image; and (2) dish-level nutritional regression estimating five aggregated nutritional quantities per dish such as calories, fats, carbohydrates, protein, and sodium.
- **Architectural Focus:** MobileNetV2 and ResNet50 are evaluated as backbone architectures, both in transfer learning configurations with ImageNet pretrained weights. Vision transformers, ensemble methods, and large-scale vision-language models are beyond the scope of this investigation but are identified as directions for future work.

1.4.2. Limitations of the Research

Every research has limitations, and acknowledging them clearly helps define the scope of the study while identifying opportunities for future improvement. The limitations of this research are summarized as follows.

- **Limitation of Image Modality:** This study utilizes only two-dimensional RGB food images as the primary input for food recognition and nutrition estimation. Consequently, the proposed framework does not have access to depth, volume, or

three-dimensional geometric information, which are important factors for accurately estimating portion-dependent nutritional values such as calories, fats, and protein. Although the proposed multi-task learning framework demonstrates promising performance, the absence of explicit volumetric information remains an inherent limitation of RGB image-based dietary assessment.

- **Dataset Size and Class Distribution:** The proposed framework was developed and evaluated using a cleaned subset of the NutritionVerse-Real dataset containing 42 food categories. Although the dataset provides valuable food images together with corresponding nutritional annotations, the number of training samples for certain food categories remains relatively limited. Furthermore, the dataset exhibits class imbalance, where some food categories are represented by substantially fewer samples than others. This imbalance may influence both classification performance and the subsequent nutrition estimation task.
- **Nutritional Targets:** This research focuses on estimating five major nutritional attributes, namely calories, fats, carbohydrates, protein, and sodium. Other important nutrients and micronutrients, such as dietary fiber, vitamins, and minerals, were not included in the proposed regression framework. Therefore, the developed system should be considered a preliminary nutritional assessment tool rather than a comprehensive nutritional analysis system.
- **Generalization Capability:** The experimental evaluation was conducted using the NutritionVerse-Real dataset, which primarily contains Asian food images. Although this dataset is appropriate for the objectives of the present study, the generalization capability of the proposed framework to other cuisines, food presentation styles, and real-world environments has not yet been extensively evaluated. Additional validation using more diverse datasets would be beneficial to further assess the robustness of the proposed approach.

Despite these limitations, the proposed framework provides a comprehensive investigation of simultaneous food recognition and nutrition estimation using multi-task learning and establishes a solid foundation for future research on intelligent dietary assessment systems.

1.5. Significant of Research

The significance of this research extends beyond the development of a deep learning model and contributes to both the academic community and practical applications of intelligent dietary assessment. The proposed framework demonstrates how multi-task learning can be effectively applied to simultaneously perform food recognition and nutrition estimation within a unified deep learning architecture. The significance of this study is summarized as follows.

- **Academic Significance:** From an academic perspective, this research contributes to the growing body of knowledge on image-based dietary assessment by proposing a classification-conditioned multi-task learning framework that integrates multi-label food recognition and nutrition estimation within a single end-to-end model. Unlike conventional approaches that treat classification and regression as separate problems, the proposed framework exploits the relationship between food identity and nutritional composition to improve learning efficiency. Furthermore, this study provides a systematic comparison between MobileNetV2 and ResNet50 under identical datasets, preprocessing procedures, training strategies, and evaluation protocols. The comparative analysis offers valuable insights into the influence of backbone architecture on both classification and regression performance within a multi-task learning framework.
- **Practical Significance:** From a practical perspective, the proposed framework demonstrates the feasibility of automatically estimating nutritional information from food images captured using standard RGB cameras. Such an approach has the potential to reduce the burden of manual dietary recording and provide a more convenient method for monitoring daily nutritional intake. In addition, the favorable balance between computational efficiency and prediction performance achieved by MobileNetV2 suggests that lightweight deep learning models can support future deployment on smartphones, mobile health applications, and other resource-constrained devices, thereby increasing the accessibility of intelligent dietary assessment technologies.
- **Societal Significance:** From a societal perspective, this research supports ongoing efforts to promote healthier dietary behaviors and improve public awareness of nutritional intake. By enabling more convenient food recognition and nutrition estimation, the proposed framework may contribute to preventive healthcare, dietary education, and lifestyle management, particularly for individuals requiring

long-term nutritional monitoring. Moreover, this study aligns with the United Nations Sustainable Development Goal 3 (SDG 3): Good Health and Well-being, which aims to reduce premature mortality from non-communicable diseases through prevention and improved health management. Although the proposed framework is intended for research purposes, it provides a technological foundation for future intelligent dietary assessment systems capable of supporting healthier and more informed food choices.

Overall, this research demonstrates the potential of multi-task deep learning for automated dietary assessment while providing practical insights into the design of efficient computer vision systems for food recognition and nutrition estimation. The findings of this study are expected to support future developments in intelligent healthcare, nutritional monitoring, and mobile health applications.

1.6. Thesis Structure

The remainder of this thesis is organized into five chapters, each addressing a distinct phase of the research from foundational review to empirical evaluation and conclusions. The structure is designed so that each chapter builds logically upon the preceding one, providing a coherent and progressive account of the work undertaken.

- **Chapter 1 - Introduction:** Establishes the research context by situating the study within the global and regional public health challenge of diet-related NCDs, with specific reference to Cambodia's nutritional transition. The chapter traces the evolution of food recognition technology from handcrafted features to deep CNN-based multi-task approaches, identifies the geographical and methodological gaps in existing research, and defines the five research objectives, scope, limitations, and original contributions of this thesis.
- **Chapter 2 - Literature Review:** Provides a systematic and comprehensive review of existing literature spanning dietary monitoring and the automation imperative, deep learning architectures for food recognition (ResNet50 and MobileNetV2), nutrition and weight estimation approaches from food images (depth-based and 2D regression methods), multi-task learning foundations and applications in food analysis, and publicly available food image datasets. The chapter critically compares related works in a structured summary table and concludes by identifying four specific research gaps in Asian multi-label recognition, 2D multi-task

nutritional estimation, classification-conditioned regression design, and per-nutrient loss weighting that directly motivate the contributions of this thesis.

- **Chapter 3 - Methodology:** Presents a detailed and fully reproducible description of the proposed framework, covering the NutritionVerse-Real dataset and its characteristics, the complete five-stage preprocessing pipeline (metadata restructuring, image validation, feature extraction, nutritional target preparation, and data splitting), the targeted augmentation and class-weighting strategies for imbalance mitigation, the shared CNN backbone architecture with classification-conditioned regression head design, the unified multi-task loss function including the novel per-nutrient weighted Huber regression loss, the two-stage transfer learning training strategy, and all evaluation metrics for both classification and regression tasks. Complete hyperparameter configurations for both MobileNetV2 and ResNet50 are tabulated to enable replication.
- **Chapter 4 - Results and Discussion:** Presents and analyses the experimental results obtained from MobileNetV2 and ResNet50 on the held-out test set. The chapter reports overall food recognition metrics (Table 4.1), analyses training dynamics through Stage 1 and Stage 2 learning curves, provides a detailed per-class F1-score breakdown identifying the strongest and weakest performing food categories (Tables 4.3 and 4.4), presents the full dish-level nutrition estimation results across all five targets (Table 4.2), and delivers a comprehensive cross-architecture comparative analysis. The discussion section interprets the key findings, validates the classification-conditioned regression mechanism, explains the 2D portion ambiguity constraint, and draws practical implications for multi-task learning design with limited training data.
- **Chapter 5 - Conclusions and Future Work:** Summarizes the key findings and original contributions of the thesis, explicitly acknowledges the identified limitations, and outlines five recommended directions for future research: depth and volume integration for portion-aware estimation, construction of a dedicated Cambodian food dataset, dynamic multi-task loss balancing, vision-language model integration, and mobile application deployment with user studies. The chapter closes with a discussion of the framework's alignment with the United Nations Sustainable Development Goals and its potential for societal impact in Cambodia and the broader Southeast Asian region.

CHAPTER 2

LITERATURE REVIEW

In this chapter, a comprehensive review of existing literature and related studies is presented to establish the theoretical and technical foundation of the research. The review begins with an exploration of dietary perceptions and the necessity of automated nutritional monitoring. Subsequently, we examine the state-of-the-art in computer vision for food analysis, specifically focusing on Deep Learning architectures for image classification and predictive models for weight and volume estimation. Furthermore, this section analyzes the evolution of specialized food datasets and the recent shift toward integrated Multi-Task Deep Learning frameworks. Finally, the chapter identifies the current research gaps in Asian cuisine modeling that this thesis aims to address.

2.1. Dietary Monitoring and the Automation Imperative

2.1.1. The Public Health Burden of Poor Dietary Habits

The relationship between dietary quality and non-communicable disease risk is among the most thoroughly established findings in epidemiology. The Global Burden of Disease Study estimates that dietary risk factors were responsible for approximately 11 million deaths and 255 million disability-adjusted life-years (DALYs) in 2017, making suboptimal diet one of the largest contributors to preventable mortality worldwide. The WHO [1] specifically identifies high sodium intake (linked to hypertension and cardiovascular disease), excessive consumption of saturated fats and trans-fats (linked to cardiovascular disease and obesity), and high free sugar intake (linked to type-2 diabetes and dental caries) as the dietary risk factors with the greatest global disease burden.

In Southeast Asia, the epidemiological transition driven by economic development and urbanization has created a double burden of malnutrition: traditional undernutrition persists in rural and peri-urban populations, while overnutrition and diet-related NCDs [2] are rapidly increasing in urban centers. Cambodia exemplifies this pattern, with MoH data indicating that overweight and obesity prevalence has nearly doubled in urban areas over the past decade, predominantly driven by increased consumption of processed foods, fast food, and sugar-sweetened beverages [1]. The direct economic cost of diet-related NCDs in terms of healthcare expenditure, lost productivity, and premature mortality is substantial and growing, creating a

compelling economic as well as public health rationale for investment in dietary monitoring technology.

2.1.2. Limitations of Traditional Dietary Assessment

Traditional dietary assessment methods each have well-documented limitations that motivate the development of automated approaches [1] [3]. The 24-hour dietary recall, in which individuals are asked to remember and report everything consumed in the preceding 24 hours, is the most widely used method in epidemiological research. However, it is subject to memory limitations, social desirability bias (under-reporting of “unhealthy” foods), and the cognitive burden of estimating portion sizes without reference standards.

Food frequency questionnaires (FFQs), while less subject to recall bias for habitual dietary patterns, require extensive validation for each cultural and dietary context and provide no information about day-to-day dietary variation [4] [7]. Food diaries improve accuracy but are associated with dietary modification during recording (“reactive recording”) and recording fatigue in longitudinal studies [4] [7]. Physical food weighing, while the most accurate method, is impractical for real-world dietary monitoring and is incompatible with communal eating styles common in Cambodia and across Southeast Asia. Biomarker-based validation studies using doubly labelled water consistently demonstrate that self-reported energy intake underestimates actual energy expenditure by 20% - 40% on average [6], confirming the systematic bias of all recall-based methods.

2.1.3. Image-Based Dietary Assessment

The concept of using food photographs as a basis for dietary assessment was formalized in the Digital Photography of Food Method (DPFM), which demonstrated that meal photographs could provide a more objective and reliable record of consumed food than retrospective recall alone. The subsequent combination of ubiquitous smartphone cameras with deep learning image recognition created the technical preconditions for fully automated image-based dietary tracking.

Early commercial implementations of food recognition in dietary tracking applications (MyFitnessPal, Lose It!) were limited to keyword-based search following user photography rather than automated recognition. Automated food recognition features were subsequently introduced in applications such as Im2Calories (Google Research, 2015), which demonstrated the feasibility of estimating calories from meal photographs using CNNs. However, the

accuracy and coverage of these systems remained limited, particularly for non-Western cuisines, motivating continued research into more capable and culturally inclusive food recognition systems.

2.2. Deep Learning Architectures for Food Recognition

2.2.1. The CNN Revolution in Food Analysis

Prior to the emergence of deep convolutional neural networks following the introduction of AlexNet [5], state-of-the-art food recognition systems relied primarily on handcrafted visual features and traditional machine learning algorithms. Although these methods achieved reasonable performance under controlled conditions, they generally exhibited limited robustness when handling large-scale datasets and visually complex food images. AlexNet’s demonstration that deep CNNs could dramatically outperform handcrafted approaches on the large-scale ImageNet benchmark sparked rapid adoption of CNN-based approaches for food recognition [5].

The key insight enabling rapid progress in food recognition was transfer learning: CNNs pre-trained on the 1.2 million images of ImageNet develop general-purpose visual feature representations in their early and middle layers that transfer effectively to specialized recognition tasks including food classification. By replacing the final classification layer and fine-tuning on domain-specific food image data, researchers could leverage millions of training images from a related visual domain to achieve strong performance even with limited food-specific training data. This transfer learning paradigm is the foundation of both architectures evaluated in this thesis.

2.2.2. ResNet50: Deep Residual Learning

He et al. [11] introduced the Residual Network (ResNet) family of architectures in their landmark 2016 CVPR paper, addressing the fundamental problem of vanishing and exploding gradients that prevented the effective training of very deep neural networks. The key innovation was the residual skip connection: a direct identity shortcut connection that adds the input of a convolutional block to its output. This allows gradients to flow directly from deeper layers to shallower layers during backpropagation without passing through the non-linear activations of the intermediate layers, effectively solving the vanishing gradient problem for networks of arbitrarily large depth.

ResNet50, the 50-layer variant, achieves 76.1% top-1 accuracy on ImageNet with approximately 25.6 million parameters. Its deep architecture provides a rich hierarchy of visual features: early layers capture low-level visual primitives (edges, colors, simple textures), middle layers encode more complex patterns and object parts, and deep layers capture high-level semantic representations of food categories and ingredients. This hierarchical depth makes ResNet50 particularly well-suited for distinguishing between visually similar food categories that share common visual elements at multiple levels of abstraction.

In food recognition research, ResNet50 is one of the most widely cited and evaluated architectures. Oguntimilehin and Toyin [12] employ ResNet50 for food category recognition in a dietary tracking system, reporting approximately 85% accuracy on their evaluation set. Queipo-Alvarez et al. [13] identify ResNet variants as among the consistently strongest performers in their comparative review of food calorie estimation models, attributing this to ResNet’s capacity for deep feature extraction and robustness to overfitting through residual connections.

2.2.3. MobileNetV2: Efficiency for Mobile Deployment

Sandler et al. [14] introduced MobileNetV2 as an evolution of the original MobileNet architecture, specifically optimized for computational efficiency on mobile and embedded devices. The architecture introduces two key innovations over standard residual networks: inverted residual blocks and linear bottlenecks.

In standard residual blocks (ResNet), the skip connection links high-dimensional representations and a bottleneck in the middle compresses them. MobileNetV2 inverts this structure: the skip connection links narrow low-dimensional bottleneck representations, while the feature processing occurs in a temporarily expanded higher-dimensional space. This inverted residual structure allows the network to leverage efficient depthwise separable convolutions, which considerably reduce computation compared with standard convolutions while maintaining gradient flow through the lightweight bottleneck connections.

The linear bottleneck design removes the non-linear ReLU activation from the output of each bottleneck layer, preserving information in the low-dimensional representations that would otherwise be destroyed by the non-linear activation. This design choice is theoretically motivated by the argument that low-dimensional manifolds cannot be effectively represented after non-linear projection, and empirically demonstrated to improve both accuracy and efficiency.

MobileNetV2 achieves 72.0% top-1 accuracy on ImageNet with approximately 3.4 million parameters and 0.3 GFLOP inference cost per image, representing a $7.5\times$ parameter reduction and $13.7\times$ computation reduction relative to ResNet50 at a cost of 4.1 percentage points in top-1 accuracy. For mobile dietary monitoring applications, this efficiency advantage is critical: MobileNetV2 can run inference in real time on mid-range smartphones without requiring cloud connectivity, enabling offline dietary monitoring in environments with limited internet access. Rokhva et al. [15] demonstrate that MobileNetV2 fine-tuned on domain-specific food images achieves accurate real-time recognition, confirming its practical suitability for food monitoring applications.

2.2.4. Multi-Label Food Recognition

Standard food recognition benchmarks are designed as single-label classification problems, where each image is assigned to exactly one of N food categories. This formulation is appropriate when images contain a single, isolated food item, but is fundamentally inadequate for real-world meal photographs that routinely contain multiple distinct food components simultaneously. A photograph of a typical Asian meal may show rice, a protein dish, a vegetable preparation, and a condiment simultaneously; assigning a single label to this image is not only inaccurate but fails to capture the information needed for nutritional estimation.

Multi-label classification addresses this limitation by allowing the model to predict the simultaneous presence of multiple binary labels independently. In deep learning implementations, multi-label classification is typically achieved by replacing the softmax activation of the output layer (which enforces mutual exclusivity between classes) with sigmoid activations (which allow independent prediction of each label), and using binary cross-entropy loss rather than categorical cross-entropy. However, multi-label classification introduces several technical challenges beyond single-label classification: the selection of per-class classification thresholds, the design of evaluation metrics that account for partial label correctness, and the handling of label imbalance across many classes simultaneously.

Queipo-Alvarez et al. [13] identify multi-label food recognition as a practically important but technically underexplored direction in food analysis research, noting that most comparative reviews focus exclusively on single-label benchmarks and that multi-label recognition frameworks for Asian food contexts are virtually absent from the published literature.

2.3. Nutrition and Weight Estimation from Food Images

2.3.1. Portion Estimation Challenge

The fundamental challenge distinguishing nutritional estimation from food recognition is that nutritional content is determined not only by what food is present but also by how much of it is consumed. For nutrients like sodium, whose concentration per unit mass is relatively consistent within a food category (e.g., soy sauce always has high sodium per gram), food type identification provides a strong signal for nutritional estimation. However, for total calories, fats, and carbohydrates, which are proportional to mass, portion size is the dominant determinant of nutritional content and is inherently ambiguous in 2D photographs.

This fundamental ambiguity has motivated three broad classes of approaches to the portion estimation problem: (1) depth-based methods that reconstruct 3D food volume from depth sensors or monocular depth estimation; (2) reference object methods that use known-size objects in the photograph (a credit card, a fork, a coin) to establish metric scale; and (3) direct regression methods that learn to predict portion-related quantities from visual features alone, relying on the statistical regularities of typical portion sizes in the training distribution. This thesis employs the third approach, which is the most practical for daily monitoring but the most constrained in achievable accuracy.

2.3.2. Depth-Based Nutritional Estimation: DPF-Nutrition

Y. Han et al. [8] represent the current state-of-the-art in depth-enhanced food nutrition estimation with their DPF-Nutrition framework, published in *Foods* (2023). DPF-Nutrition employs a two-stage approach: first, a monocular depth estimation network predicts a dense pixel-wise depth map from a standard 2D food image; second, the predicted depth map is fused with the original RGB image features through a dedicated RGB-D fusion module to jointly estimate nutritional values. The depth map provides volumetric cues about the three-dimensional extent of food portions, enabling the model to partially compensate for the portion ambiguity inherent in 2D-only approaches.

DPF-Nutrition demonstrates a mean Percentage Mean Absolute Error (PMAE) improvement of approximately 2.1% over comparable RGB-only models, confirming that even imperfect monocular depth estimates provide useful supplementary information for portion-aware nutritional estimation. The framework is an important existence proof that depth

information benefits nutritional estimation even when depth itself must be predicted from 2D images.

However, DPF-Nutrition has several practical limitations. The monocular depth estimation module introduces an additional source of prediction error that propagates through the nutritional estimation pipeline; depth prediction is particularly unreliable for flat dishes, liquid foods, and surfaces with specular reflections (glossy sauces, wet ingredients). The framework also requires depth estimation models pre-trained on general scene depth datasets, which may not capture the specific geometric characteristics of food scenes. The additional computational overhead of depth estimation limits real-time mobile deployment. These considerations motivate the 2D-only approach adopted in this thesis.

2.3.3. 2D Regression Approaches

As an alternative to depth-based methods, Wimalasiri and Sahoo [9] propose a 2D-only nutritional estimation framework in which a regression head built on a MobileNetV3 backbone learns to predict food weight directly from visual features and bounding box metadata including aspect ratio and average pixel intensity. This approach demonstrates that useful nutritional estimates can be obtained from 2D images without depth sensors by leveraging statistical regularities between food visual appearance and typical portion sizes in the training distribution.

The integration of global image features (aspect ratio, average pixel intensity) as auxiliary inputs to the regression head is particularly relevant to this thesis, which adopts an equivalent strategy to provide low-cost visual priors about dish composition. The key architectural innovation proposed in this thesis beyond Wimalasiri and Sahoo [9] is the explicit conditioning of the regression head on food classification outputs, which provides compositional priors beyond what can be inferred from global image features alone.

2.4. Multi-Task Learning

2.4.1. MTL in Food Analysis

He et al. [16] provided an early demonstration of multi-task learning for food analysis by jointly training food classification and portion size estimation within a shared CNN, demonstrating that jointly trained models consistently outperform separately trained single-task baselines on both tasks. Their work established the foundational evidence for the

complementary nature of food classification and nutritional estimation and motivated subsequent research into integrated multi-task food analysis frameworks.

The framework proposed in this thesis builds upon this foundation with two key architectural innovations. First, rather than passing identical shared backbone features to both branches, the proposed framework implements a classification-conditioned regression design: the regression head receives food composition predictions (classification hidden activations and output probabilities) as explicit additional inputs, creating directed information flow from classification to regression. Second, the proposed per-nutrient weighted Huber regression loss addresses the heterogeneous predictability of the nutritional targets, a challenge that has received limited attention in previous multi-task food-analysis studies.

2.4.2. Loss Function Design in MTL

The design of the combined loss function is a critical practical challenge in multi-task learning. When task losses have very different magnitudes, gradient magnitudes, or convergence rates, naive equal-weight summation can cause one task to dominate training at the expense of others. Several principled approaches have been proposed: GradNorm [17] dynamically adjusts task weights based on gradient norm ratios.

In multi-task learning, the overall optimization objective is commonly formulated as a weighted combination of the individual task losses. One straightforward approach is to assign fixed weights to each task based on their relative importance. More recently, uncertainty-weighted multi-task learning proposed by Kendall et al. automatically learns task weights by modelling task-dependent (homoscedastic) uncertainty, allowing the network to adaptively balance classification and regression losses during training rather than relying on manually selected weighting coefficients [18]. In this thesis, a fixed weighting scheme ($w_1 = 0.70$ for food recognition and $w_2 = 0.30$ for nutrition estimation) was adopted. This decision was motivated by the architectural design of the proposed framework, in which the classification branch provides semantic information that is subsequently incorporated into the regression branch. Consequently, accurate food recognition was considered a prerequisite for improving nutrition estimation performance. Although fixed weighting produced stable convergence and satisfactory overall performance, dynamic loss balancing methods such as uncertainty-weighted optimization remains a promising direction for future research [18].

2.5. Food Image Datasets

2.5.1. Overview of Major Datasets

Table 2.1 provides a comparative summary of publicly available food image datasets relevant to food recognition and nutrition estimation research.

Table 2.1: Comparative Analysis and Overall Discussion

Dataset	Year	Classes	Images	Region	Key Characteristics
Food-101 [19]	2014	101	101,000	Western	Single-label; no nutritional metadata; dominant benchmark
UECFood-100 [20]	2014	100	14,361	Japanese	Single-label; Japanese cuisine focus; web-scraped
VIREO Food-172 [21]	2016	172	110,241	Chinese	Single-label; Chinese cuisine; large scale
FoodLog [22]	2017	85	~30,000	Japanese	Mobile captured; longitudinal; single-label
UEC Food-256 [23]	2016	256	31,395	Japanese	Single-label; larger UEC variant
NutritionVerse-Real [10]	2023	45	~1,004	Asian	Multi-label; nutritional metadata; 2D; used in this thesis

2.5.2. The NutritionVerse-Real Dataset

The NutritionVerse-Real dataset [10] was introduced by Tai et al. as a resource specifically designed for dietary intake estimation research, and is unique among publicly available datasets in providing 2D food images of real dishes paired with both multi-label class annotations and comprehensive dish-level nutritional metadata. The dataset was collected by photographing real dishes prepared and served in a controlled environment, with each dish photographed from four viewpoints to capture visual diversity.

The dataset provides several attributes essential for the research conducted in this thesis. Each dish has up to seven ingredient-level food type annotations drawn from a vocabulary of 45 unique food types, directly enabling multi-label classification training. Dish-level nutritional metadata includes calories (kCal), fat (g), carbohydrates (g), protein (g), calcium (mg), iron (mg), magnesium (mg), potassium (mg), sodium (mg), vitamin D (μg), and vitamin B12 (μg), providing comprehensive macronutrient and micronutrient coverage. The multi-viewpoint photography per dish enables data augmentation through view sampling.

However, the dataset has important limitations directly relevant to this research. The total scale of 251 dish identities is modest for deep learning, particularly given the 45-category multi-label label space. The class distribution is severely imbalanced, with some categories having over 50 dish instances and others having fewer than 5. The food categories reflect North American adaptations of Asian dishes rather than authentic Southeast Asian cuisine, limiting direct applicability to the Cambodian dietary context that motivates this research.

2.5.3. Addressing Class Imbalance in Food Datasets

Class imbalance is a pervasive challenge in food recognition datasets, arising from the natural long-tail distribution of food popularity (common foods are photographed far more frequently than rare ones) and the practical difficulty of collecting images for regional specialty dishes. Adegbenjo and Ngadi [24] provide a systematic analysis of imbalance handling strategies in agri-food data analysis, identifying three primary approaches: resampling (oversampling minority classes, undersampling majority classes), cost-sensitive learning (class-weighted losses that penalize minority class misclassification more heavily), and data augmentation (generating synthetic additional training samples from existing ones through image transformations).

Their analysis finds that hybrid strategies combining augmentation with cost-sensitive learning consistently outperform single-strategy approaches, and that the specific choice of augmentation transformations should be domain-appropriate transformations that alter the identity-determining features of food images (such as extreme color shifts or spatial distortions) should be avoided. In this thesis, a hybrid strategy combining targeted augmentation (horizontal flipping, mild rotation, brightness/color jitter) with per-class positive-weighted BCE loss is adopted, consistent with the best practices identified by Adegbenjo and Ngadi [24].

2.6. Related Works Summary

Table 2.2 provides a structured comparative summary of the most relevant prior works relative to this thesis.

Table 2.2: Summary of Related Works in Food Recognition and Nutrition Estimation

Reference	Method	Dataset	Task	Key Gap vs. This Thesis
Oguntimilehin & Toyin [12]	ResNet50, CNN	Proprietary	Classification	Single-label only; no nutrition; no Asian food

Han et al. [8]	DPF-Nutrition depth+RGB	Proprietary	Nutrition est.	Requires depth; no classification; not Asian
Wimalasiri & Sahoo [9]	MobileNetV3 regression	Proprietary	Weight est.	No classification branch; single-task; not Asian
He et al. [16]	MTL CNN classification + portion	Proprietary	Cls + portion	No multi-label; no Asian food; no regression
Queipo-Alvarez et al. [13]	Review: CNNs	Multiple	Cls + calorie review	Review only; limited Asian; no multi-label
Tai et al. [10]	Dataset paper	NutritionVerse -Real	Dataset only	No model proposed; dataset introduction only
This research	MTL: MNV2/RN50, cond. regression	NutritionVerse -Real	Multi-label cls + nutrition reg.	2D only; limited dataset size

2.7. Research Gap and Positioning

The review presented in this chapter establishes four clear and interconnected research gaps that motivate the contributions of this thesis:

- **Asian Multi-Label Recognition with Nutritional Data:** No published work has developed or evaluated a multi-label food recognition framework targeting Asian food datasets with paired nutritional metadata in a multi-task architecture. The gap exists both in terms of data resources and model development.
- **2D-Only Multi-Task Nutritional Estimation:** While depth-based estimation methods [16] demonstrate the value of spatial information, no unified multi-task framework simultaneously performs multi-label food recognition and direct nutritional regression from 2D-only images on an Asian food dataset.
- **Classification-Conditioned Regression Architecture:** Existing MTL food frameworks use parallel branches sharing backbone features without explicit information flow from classification to regression. The design in this thesis where classification probabilities and hidden activations are concatenated as regression inputs has not been evaluated in this domain.
- **Per-Nutrient Loss Weighting:** Existing nutritional regression approaches treat all targets uniformly in the loss function despite their substantially different

predictabilities from 2D images. The per-nutrient weighted Huber loss proposed here is a novel approach to improving training signal quality for heterogeneously predictable regression targets.

This thesis addresses all four gaps through a unified experimental framework on the NutritionVerse-Real dataset, providing both architectural contributions and empirical evaluation evidence.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology adopted to develop and evaluate the proposed image-based nutrition estimation framework for Asian food using 2D food images. The research methodology consists of six major stages, namely data acquisition, data preprocessing, data augmentation, model development, model training, and performance evaluation. A multi-task deep learning framework was designed to simultaneously perform multi-label food recognition and nutrition estimation by sharing visual representations between both tasks. Two transfer learning models, MobileNetV2 and ResNet50, were implemented and compared to investigate the trade-off between computational efficiency and predictive performance.

3.1. Research Framework

This research adopted a quantitative experimental approach to develop and evaluate an end-to-end deep learning framework capable of simultaneously recognizing food items and estimating nutritional values from a single RGB food image. Fig.1 illustrates the overall research framework adopted throughout this study.

The framework consists of six sequential phases. The first phase involves selecting and preparing the NutritionVerse-Real dataset, followed by comprehensive preprocessing to ensure data consistency and quality. The cleaned dataset is then subjected to image augmentation to increase data diversity and improve model generalization. Subsequently, two convolutional neural network (CNN) backbones, namely MobileNetV2 and ResNet50, are integrated into a multi-task learning architecture to perform both food recognition and nutrition estimation simultaneously. Finally, the trained models are evaluated using classification and regression performance metrics to determine their effectiveness. Unlike conventional single-task approaches, the proposed framework allows both tasks to share a common feature extractor, enabling the nutrition estimation branch to leverage semantic information learned from food recognition. This shared representation is expected to improve learning efficiency while reducing computational complexity.

This study adopted a quantitative experimental approach to develop and evaluate a multi-task deep learning framework for simultaneous food recognition and nutrition estimation from 2D food images. The overall research methodology consists of six major stages,

beginning with dataset preparation and ending with model performance evaluation. Figure 1 illustrates the complete research methodology framework adopted in this study.

The proposed methodology was designed to ensure a systematic workflow for data preparation, model development, training, and evaluation while enabling a fair comparison between MobileNetV2 and ResNet50 under identical experimental settings. The framework comprises six sequential stages: (1) dataset preparation, (2) data preprocessing, (3) data augmentation, (4) model development, (5) model training, and (6) model evaluation. Each stage is briefly described as follows in Figure 3.1.

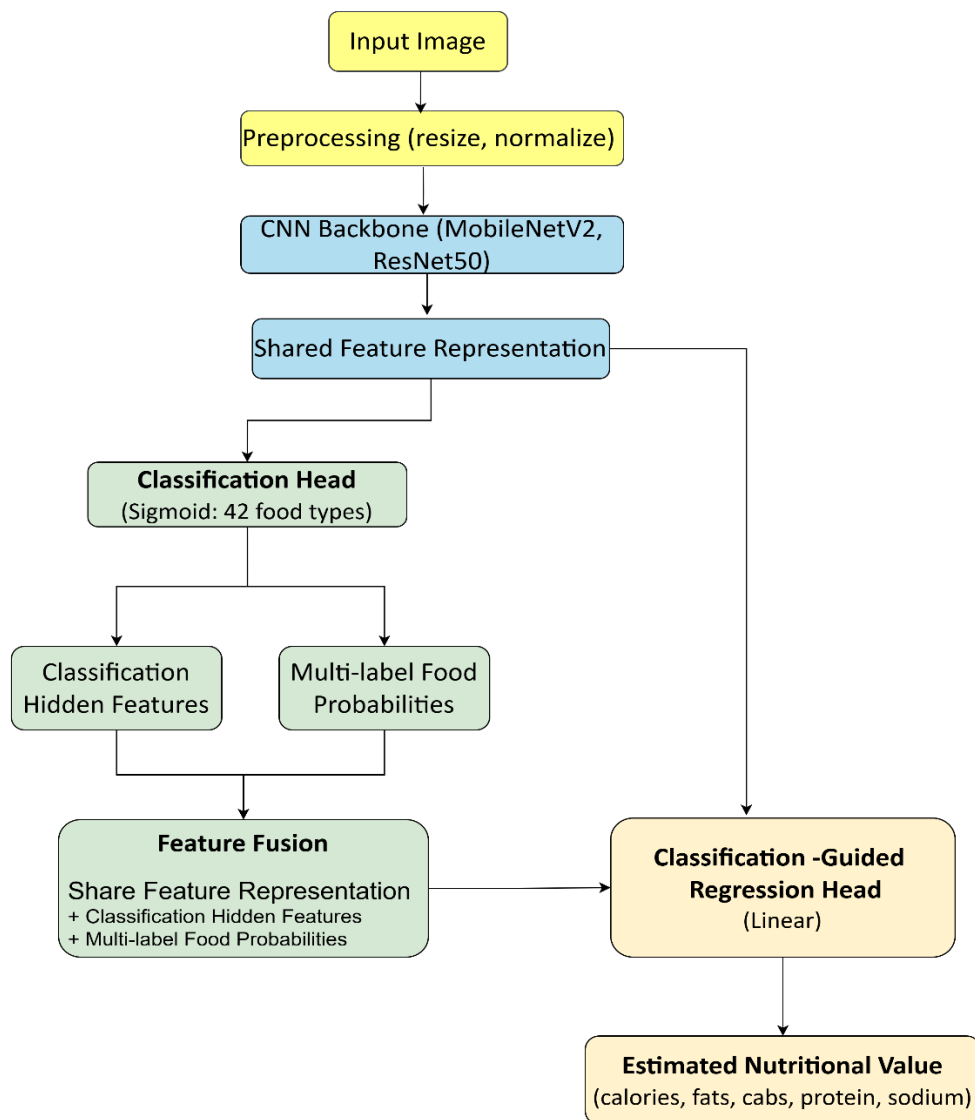


Figure 3.1: The proposed flowchart for food image analysis and nutritional estimation

- **Stage 1 - Dataset Preparation:** The first stage involved selecting the NutritionVerse-Real dataset as the primary data source. This dataset was chosen because it provides both food images and comprehensive nutritional metadata, making it suitable for developing a multi-task learning framework that performs food recognition and nutrition estimation simultaneously. Since this study focuses on Asian cuisine, only the selected Asian food categories and five nutritional targets (calories, fats, carbohydrates, protein, and sodium) were retained for subsequent analysis.
- **Stage 2 - Data Preprocessing:** The second stage focused on preparing the dataset for model development. Metadata cleaning was performed to remove inconsistent records, standardize nutritional information, and retain only the required variables. Food labels were converted into multi-hot encoded vectors to support multi-label classification, while nutritional values were extracted as continuous regression targets. In addition, all food images were resized to 224×224 pixels and normalized according to the preprocessing requirements of the pretrained CNN backbones. Finally, the processed dataset was divided into training, validation, and testing subsets.
- **Stage 3 - Data Augmentation:** To improve model generalization and reduce overfitting caused by limited training samples, target data augmentation was applied exclusively to the training dataset. Various image transformation techniques, including random horizontal flipping, rotation, zooming, translation, and brightness adjustment, were employed to simulate variations commonly encountered in real-world food photography.
- **Stage 4 - Model Development:** The fourth stage involved developing the proposed multi-task learning framework using two transfer learning architectures which are MobileNetV2 and ResNet50. In both models, the pretrained convolutional backbone served as a shared feature extractor for two parallel prediction branches. The first branch performs multi-label food recognition, while the second branch estimates five nutritional values using both the extracted visual features and the predicted food composition. This classification-conditioned regression strategy enables the regression branch to leverage semantic information obtained from food recognition, thereby improving nutritional prediction capability.

- **Stage 5 - Model Training:** The proposed models were trained using a two-stage transfer learning strategy. During the first stage, the pretrained backbone was frozen, allowing only the classification and regression heads to learn task-specific representations. In the second stage, the last convolutional layers of the backbone were selectively unfrozen and fine-tuned using a lower learning rate to adapt the pretrained features to the food image domain. To optimize both learning objectives simultaneously, the model employed a combined loss function consisting of Weighted Binary Cross-Entropy (WBCE) for multi-label classification and Huber Loss for multi-output regression.
- **Stage 6 - Model Evaluation:** The final stage evaluated the performance of both architectures using classification and regression metrics. The food recognition task was assessed using Precision, Recall, F1-score, Macro F1-score, Micro F1-score, Area Under the ROC Curve (AUC), Hamming Accuracy, and Hamming Loss. Meanwhile, the nutrition estimation task was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). The performance of MobileNetV2 and ResNet50 was then compared under identical experimental conditions to identify the architecture that provides the most effective balance between food recognition accuracy and nutrition estimation performance.

3.2. Dataset Description

3.2.1. NutritionVerse-Real Dataset

This study utilized the NutritionVerse-Real dataset developed by Tai et al. [10], which is one of the largest publicly available datasets specifically designed for image-based dietary assessment. Unlike conventional food recognition datasets that provide only class labels, NutritionVerse-Real includes both visual information and detailed nutritional metadata, making it particularly suitable for multi-task learning applications involving food recognition and nutrition estimation. The original dataset contains manually collected food images representing various cuisines together with nutritional annotations obtained from standardized nutritional databases. Each image is associated with dish-level nutritional information, including calories, fats, carbohydrates, protein, sodium, vitamins, minerals, serving weight, and food composition labels. The availability of both food labels and nutritional information

enables the development of an integrated framework capable of performing simultaneous food recognition and nutrition estimation from a single image.

3.2.2. Dataset Characteristics

Although NutritionVerse-Real contains diverse food categories collected from different cuisines, this research focused exclusively on Asian food to better align with the dietary habits commonly observed in Cambodia and neighboring countries. The NutritionVerse-Real is an open access manually collected 2D food scene dataset for dietary intake estimation with 889 images of 251 distinct dishes and 45 unique food types which contains real Asian dishes captured from multiple angles which each dish id have 4 images corresponding with nutritionverse_dish_metadata3 file which included dish_id, food_item_type, food_weight(g), calories(kCal), fat(g), carbohydrates(g), protein(g), calcium(mg), iron(mg), magnesium(mg), potassium(mg), sodium(mg), vitamin_d(μ g), vitamin_b12(μ g) [10]. Figure 3.2 provides details of the food class distribution of the original data.

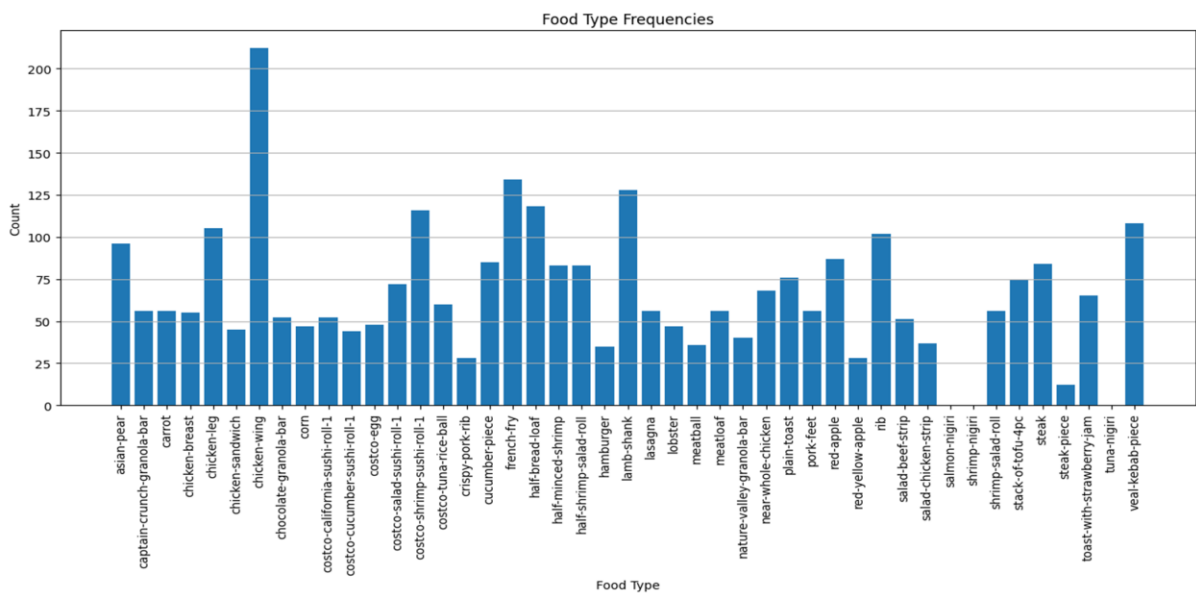


Figure 3.2: Food-class Distribution of Original Data

Figure 3.3 provides the example dishes from NutritionVerse-Real datasets.

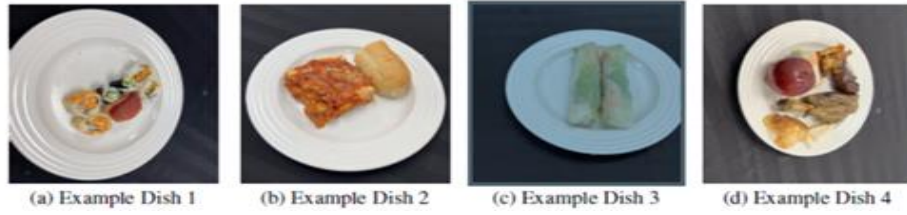


Figure 3.3: Example dishes from NutritionVerse-Real

The five nutritional targets selected in this study include Calories (kcal), Fat (g), Carbohydrates (g), Protein (g), Sodium (mg). These nutrients were selected because they are commonly used in dietary assessment and are directly related to major non-communicable diseases such as obesity, hypertension, cardiovascular disease, and diabetes.

3.2.3. Food Category Selection

The original NutritionVerse-Real dataset contains hundreds of individual food components and dishes. However, many categories contain very few samples, which can negatively affect multi-label classification performance. To improve model reliability, only 42 commonly occurring Asian food categories with sufficient training samples were selected. The selected food categories represent a diverse range of staple foods, vegetables, fruits, meats, seafood, and processed foods commonly consumed in Asian countries. Restricting the study to these food categories reduces extreme class imbalance while maintaining sufficient diversity for evaluating the proposed framework under realistic dietary scenarios.

3.3. Data Preprocessing

Data preprocessing is a crucial step in deep learning applications because the quality of the input data directly influences model performance. Raw datasets often contain inconsistencies, missing values, duplicated records, and variations in image quality that may negatively affect the learning process. Therefore, several preprocessing procedures were performed to ensure that both the image data and nutritional metadata were properly prepared before model training.

3.3.1. Metadata Cleaning

The NutritionVerse-Real dataset provides detailed nutritional metadata together with food images. However, the original metadata contains numerous nutritional attributes, ingredient descriptions, serving information, and food identifiers that are not directly required

for this study. Therefore, metadata cleaning was conducted to extract only the variables relevant to the proposed multi-task learning framework.

Several preprocessing operations were performed, including the removal of duplicated records, verification of missing nutritional values, standardization of food labels, and selection of the five nutritional targets used in this study, namely calories, fats, carbohydrates, protein, and sodium. In addition, ingredient-level annotations were aggregated into dish-level nutritional records so that each food image corresponded to a single nutritional label. This process ensured consistency between the image data and their associated nutritional information while reducing potential sources of noise during model training.

To ensure the reliability of the training process, a systematic filtering step was performed to verify the presence of valid image references for every dish in the metadata. Images were loaded and verified using the Python Imaging Library (PIL); dishes lacking valid images were excluded to prevent mismatches. After cleaning incomplete classes and missing metadata, the study obtained 889 original images across 42 food classes. The final dataset includes multi-label annotations representing ingredient presence, along with comprehensive nutritional metadata including food weight (g), calories (kCal), fat (g), carbohydrates (g), protein (g), and various micro-nutrients such as calcium, iron, and vitamins. During this process, three specific classes with insufficient data; salmon-nigiri, shrimp-nigiri, and tuna-nigiri were removed to maintain statistical significance. Figure 3.4 provides details of the food class distribution after dropping missing matches and refining the metadata.

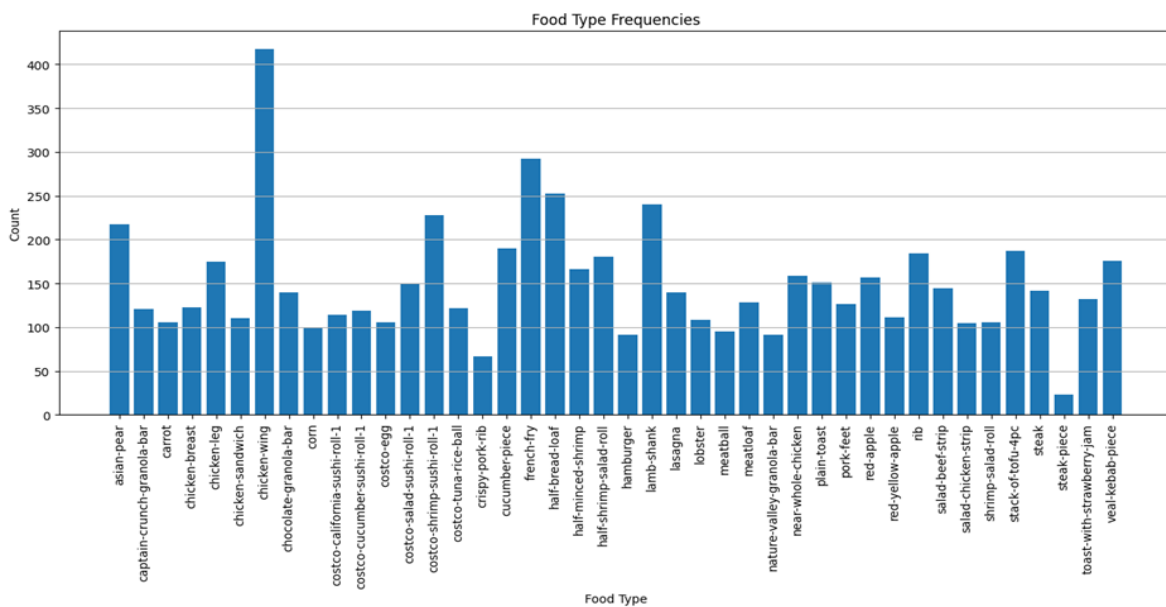


Figure 3.4: Food-class Distribution After Removing Unmatched Records.

3.3.2. Image Preprocessing

Image preprocessing was performed to standardize the visual inputs before they were fed into the convolutional neural networks. Since the original images in the NutritionVerse-Real dataset have different resolutions and aspect ratios, all images were resized to 224×224 pixels, which corresponds to the default input size required by both MobileNetV2 and ResNet50. Following image resizing, pixel values were normalized using the preprocessing functions provided by the respective pretrained models. This normalization aligns the input image distribution with that of the ImageNet dataset on which the backbone networks were originally trained, thereby improving transfer learning effectiveness and accelerating model convergence. Furthermore, corrupted or unreadable image files were identified and excluded from the dataset to ensure that all training samples were valid and could be processed successfully during model training.

3.3.3. Label Encoding

The food recognition task was formulated as a multi-label classification problem because a single dish may contain multiple food components simultaneously. Consequently, conventional one-hot encoding is not suitable for representing the food labels. Instead, multi-hot encoding was adopted, where each of the 42 food categories was represented by a binary indicator. A value of 1 indicates that a particular food item is present in the dish, while 0 indicates its absence. This encoding strategy enables the model to predict multiple food categories simultaneously and provides label-conditioned information for the subsequent nutrition estimation task. In addition to the classification labels, the corresponding nutritional values were extracted and organized into a five-dimensional continuous target vector consisting of calories, fats, carbohydrates, protein, and sodium. These nutritional targets form the regression outputs of the proposed multi-task learning framework.

3.3.4. Dataset Splitting

To ensure a fair and unbiased evaluation of the proposed multi-task learning framework, the cleaned dataset was divided into training, validation, and testing subsets before model training. The dataset partitioning was performed based on the unique dish_id rather than individual food images to prevent data leakage. This strategy ensures that images belonging to the same dish are assigned to only one subset, preventing visually similar samples from appearing in both the training and testing datasets. The original dataset consisted of 889 food images, which were divided into 711 training images (80%), 88 validation images (10%), and 90 testing images (10%). The validation set was used for hyperparameter tuning, model

selection, and early stopping, while the testing set remained completely unseen throughout the training process and was used exclusively for the final model evaluation. After dataset splitting, targeted data augmentation was applied only to the training subset to alleviate class imbalance and improve model generalization. A total of 62 augmented images were generated for minority food classes, increasing the training set from 711 to 773 images. No augmentation was applied to the validation or testing subsets, which remained at 88 and 90 images, respectively. This strategy ensures that model evaluation is performed on authentic, unseen images and avoids information leakage from augmented samples.

3.4. Data Augmentation

Deep learning models generally require a large amount of diverse training data to achieve strong generalization performance. However, collecting and annotating food images with detailed nutritional information is both labor-intensive and costly. Moreover, several food categories in the NutritionVerse-Real dataset contain relatively limited training samples, increasing the risk of overfitting. To improve the diversity of the training data and enhance model robustness, online data augmentation was applied exclusively to the training set. During each training iteration, images were randomly transformed while preserving their semantic content. The augmentation techniques employed in this study include horizontal flipping, random rotation, random zooming, random translation, and brightness adjustment.

These transformations simulate common variations encountered in real-world food photography, such as different camera viewpoints, lighting conditions, image orientations, and food placements. Consequently, the model is encouraged to learn more invariant and discriminative visual features, thereby improving its ability to generalize to unseen images. To investigate the effectiveness of data augmentation, two experimental settings were evaluated: without augmentation and with augmentation. This comparison enables an assessment of the influence of augmentation on both food recognition and nutrition estimation performance, reported later in this chapter alongside the corresponding classification and regression results.

Targeted augmentation was applied only to minority food classes in the training split, defined as classes with fewer than 30 training images. For each such class, additional images were generated from its existing training images (horizontal flip, small-angle rotation, and brightness/colour jitter) up to a target of 30 images per class, with each source image reused at most 3 times to limit duplication. This increased the training set from 711 to 773 images, an increase of 62 images (8.7%). Table 3.1 summarises the classes affected. Three additional

minority classes (salmon-nigiri, tuna-nigiri, and shrimp-nigiri) had zero training images and could therefore not be augmented, since no source image was available to generate variations from. The class-level totals above sum to 59 additional images; the realized total of 62 reflects minor variation introduced by the per-image duplication cap during generation.

Table 3.1: Training Classes Affected by Targeted Augmentation

Food Class	Images Before	Target per Class	Images Added
red-yellow-apple	8	30	22
steak-piece	16	30	14
hamburger	23	30	7
lasagna	24	30	6
chicken-sandwich	26	30	4
crispy-pork-rib	28	30	2
meatloaf	28	30	2
chocolate-granola-bar	28	30	2

3.5. Proposed Multi-Task Learning Framework

These Recent advances in deep learning have demonstrated that multiple related tasks can be learned simultaneously through a shared neural network architecture, a paradigm commonly known as multi-task learning (MTL). Unlike conventional single-task learning, where each task is optimized independently, MTL enables several learning objectives to share common feature representations. By exploiting the intrinsic relationship between related tasks, MTL has been shown to improve model generalization, reduce overfitting, and enhance learning efficiency, particularly when training data are limited. In the context of dietary assessment, food recognition and nutrition estimation are inherently related tasks. The nutritional composition of a dish is strongly influenced by the types of food ingredients it contains. Therefore, accurately identifying the food components can provide valuable semantic information that supports the estimation of nutritional values. Motivated by this relationship, this study proposes an end-to-end multi-task learning framework that simultaneously performs multi-label food recognition and nutrition estimation from a single 2D food image.

Unlike traditional approaches that develop separate models for food recognition and nutrition prediction, the proposed framework integrates both tasks into a unified architecture with a shared feature extraction backbone. This design allows the model to learn generalized visual representations that are beneficial to both tasks while reducing computational redundancy and improving training efficiency. The proposed framework consists of three major components which first a shared feature extraction backbone, second multi-label food classification branch and finally nutrition estimation regression branch.

As shown in Figure 3.1, an input food image is first processed by a pretrained convolutional neural network (CNN) backbone to extract high-level visual representations. These shared features are subsequently forwarded to two parallel prediction branches. The first branch predicts the presence or absence of multiple food categories within the image, while the second branch estimates five nutritional values, namely calories, fats, carbohydrates, protein, and sodium. Unlike conventional multi-task architectures that rely solely on shared visual features, the proposed framework further incorporates the predicted food composition into the nutrition estimation branch. Specifically, the hidden feature representation and the classification output probabilities are concatenated with the shared visual features before being passed to the regression network. This classification-conditioned regression strategy enables the nutrition estimation branch to leverage semantic information regarding the predicted food ingredients, thereby improving its ability to infer nutritional content from visual information.

Overall, the proposed architecture aims to establish a more effective relationship between food recognition and nutrition estimation, allowing both tasks to complement one another while maintaining a single end-to-end learning framework.

3.5.1. Overall Architecture

The proposed framework adopts a hard parameter sharing multi-task learning architecture, where the convolutional backbone is shared between the classification and regression tasks. Hard parameter sharing is one of the most widely used multi-task learning strategies because it enables multiple tasks to utilize common feature representations while maintaining separate task-specific prediction layers. In this study, the backbone network serves as a universal feature extractor that transforms the input food image into a compact high-dimensional feature representation. These shared features capture important visual characteristics such as texture, color, shape, and spatial arrangement of food items, which are useful for both food recognition and nutrition estimation.

This architecture allows both tasks to be optimized simultaneously during training using a combined loss function, enabling the shared backbone to learn representations that are beneficial to both objectives. Compared with training two independent models, the proposed framework offers several advantages:

- It reduces computational redundancy by sharing a common feature extractor.
- It enables both tasks to learn complementary visual representations.
- It improves parameter efficiency by reducing the number of trainable parameters.

- It allows semantic information learned during food recognition to support nutrition estimation.
- It simplifies deployment by producing both food recognition and nutritional predictions within a single inference pipeline.

These advantages make multi-task learning particularly suitable for dietary assessment systems, where food identification and nutritional estimation are closely related prediction tasks.

3.5.2. Shared Feature extraction

The first component of the proposed framework is the shared feature extraction backbone. Convolutional Neural Networks (CNNs) have demonstrated outstanding performance in numerous computer vision applications due to their ability to automatically learn hierarchical visual representations from raw image data. Rather than relying on handcrafted features, CNNs progressively extract low-level and high-level image features through multiple convolutional layers.

In this study, the backbone network receives a preprocessed food image with an input size of $224 \times 224 \times 3$. The image is passed through a series of convolutional operations to generate feature maps that encode discriminative visual characteristics of the food, including texture, color distribution, object boundaries, and spatial structure. Instead of training a CNN from scratch, transfer learning was adopted by initializing the backbone with pretrained ImageNet weights. This approach enables the network to leverage previously learned generic visual features while adapting to the food image domain through fine-tuning. The use of transfer learning not only accelerates convergence but also improves generalization performance, particularly when the available training dataset is relatively limited.

To investigate the influence of backbone architecture on multi-task learning performance, two widely adopted CNN models were evaluated in this study:

- **MobileNetV2:** representing a lightweight and computationally efficient architecture suitable for resource-constrained environments.
- **ResNet50:** representing a deeper architecture with stronger feature extraction capability through residual learning.

Although the two backbones differ in architectural complexity, both generate a shared feature representation that is subsequently utilized by the classification and regression branches. This design ensures that the comparison between MobileNetV2 and ResNet50

focuses primarily on the effectiveness of the backbone while maintaining identical downstream prediction heads.

3.5.3. Multi-Label Classification Branch

The first prediction task of the proposed framework is multi-label food recognition. Unlike conventional image classification problems where each image belongs to a single class, a food dish frequently contains multiple ingredients simultaneously. Consequently, the food recognition task was formulated as a multi-label classification problem, allowing the model to predict multiple food categories from a single image.

The classification branch receives the shared feature representation extracted by the backbone network and passes it through a sequence of fully connected layers. Batch normalization and dropout layers are incorporated to stabilize training and reduce overfitting, while nonlinear activation functions enable the network to learn more complex feature relationships. The final output layer consists of 42 neurons, corresponding to the selected food categories used in this study. Each neuron employs a sigmoid activation function, producing an independent probability ranging from 0 to 1 that indicates the likelihood of the corresponding food category being present in the input image.

Unlike the softmax activation commonly used in single-label classification, sigmoid activation allows multiple output neurons to be activated simultaneously, making it more appropriate for multi-label food recognition. This design enables the model to recognize multiple food components within a single dish, thereby providing richer semantic information for the subsequent nutrition estimation task. In addition to generating the final classification probabilities, the intermediate hidden feature representation learned by the classification branch is also utilized by the regression branch. This hidden representation captures higher level semantic relationships among food categories and serves as an additional source of information for nutrition estimation.

3.5.4. Classification-Conditioned Regression Branch

The second prediction task is dish-level nutrition estimation. Rather than estimating nutritional values solely from visual features, the proposed framework incorporates information obtained from the food recognition task to guide the regression process. Specifically, the regression branch receives three sources of information:

- The shared visual features extracted by the CNN backbone.
- The hidden feature representation from the classification branch.

- The predicted food category probabilities produced by the classification branch.

These three feature sources are concatenated into a unified feature vector before being passed through a series of fully connected regression layers to estimate the five nutritional targets: calories, fats, carbohydrates, protein, and sodium. This design is referred to as classification-conditioned regression, as the regression task is explicitly conditioned on the food recognition results. The rationale behind this approach is that nutritional values are inherently associated with the types of food present in a dish. For example, two visually similar dishes may have substantially different nutritional compositions if they contain different ingredients. By incorporating food recognition outputs into the regression process, the model can utilize semantic information that may not be fully captured by visual features alone.

The regression branch produces five continuous outputs simultaneously, corresponding to the selected nutritional targets. During training, these outputs are optimized using Huber Loss, while the classification branch is optimized using Weighted Binary Cross-Entropy. Both tasks are jointly optimized through a combined loss function, enabling the shared backbone to learn representations that benefit both food recognition and nutrition estimation.

3.6. Backbone Networks

The performance of deep learning models in computer vision tasks is highly dependent on the effectiveness of the feature extraction network, commonly referred to as the backbone network. In image-based food analysis, the backbone is responsible for learning hierarchical visual representations that capture discriminative characteristics such as color, texture, shape, and spatial arrangement of food items. These learned representations form the foundation for subsequent prediction tasks, including food recognition and nutrition estimation.

Training a deep convolutional neural network from scratch generally requires a very large amount of annotated data and substantial computational resources. However, publicly available food nutrition datasets remain relatively limited compared with large-scale image datasets such as ImageNet. Consequently, training an entirely new deep neural network may lead to slow convergence and poor generalization due to insufficient training samples.

To address this limitation, this study adopts transfer learning, which enables the proposed framework to leverage knowledge learned from large-scale image datasets while adapting the model to the food image domain through fine-tuning. Two widely adopted convolutional neural network architectures, namely MobileNetV2 and ResNet50, were selected as the shared feature extraction backbones. These architectures represent two different

design philosophies: MobileNetV2 emphasizes computational efficiency through lightweight convolutional operations, whereas ResNet50 focuses on learning deeper feature representations using residual learning. Comparing these two architectures allows the study to investigate the trade-off between computational efficiency and predictive performance within the proposed multi-task learning framework.

3.6.1. Transfer Learning

Transfer learning has become one of the most widely adopted techniques in deep learning, particularly when the available training data are insufficient for training deep neural networks from scratch. Instead of randomly initializing all model parameters, transfer learning begins with a network that has already learned generic visual features from a large benchmark dataset, allowing these learned representations to be transferred to a new application domain. In this research, both MobileNetV2 and ResNet50 were initialized using weights pretrained on the ImageNet dataset, which contains more than one million natural images across one thousand object categories. During pretraining, these networks learn hierarchical visual features that range from simple edge detectors in early convolutional layers to complex semantic representations in deeper layers. Although ImageNet does not specifically contain food nutrition labels, many of these low-level and mid-level visual features remain transferable to food image analysis.

The transfer learning process adopted in this study consists of two sequential stages. During the first stage, all backbone layers were frozen, allowing only the task-specific classification and regression heads to be optimized. This enables the newly added prediction layers to learn meaningful mappings while preserving the pretrained visual representations. Once the prediction heads had stabilized, the second stage involved selectively unfreezing the deeper convolutional layers of the backbone and jointly optimizing the entire network using a smaller learning rate. Fine-tuning allows the pretrained features to gradually adapt to domain-specific characteristics present in food images while minimizing the risk of catastrophic forgetting.

The two-stage fine-tuning strategy provides several advantages. First, it reduces the number of trainable parameters during the initial training phase, thereby accelerating convergence. Second, it improves model stability by preventing large weight updates to pretrained layers before the prediction heads have learned reasonable representations. Finally,

gradual fine-tuning enables the shared backbone to learn food-specific visual patterns while retaining the robust feature representations obtained from ImageNet pretraining.

3.6.2. MobileNetV2

MobileNetV2 was selected as the first backbone network because it provides an effective balance between classification accuracy and computational efficiency. Developed by Sandler et al. [14], MobileNetV2 was specifically designed for mobile and embedded devices where computational resources and memory are limited. Despite its lightweight architecture, MobileNetV2 has demonstrated competitive performance across numerous image classification and object recognition tasks. Unlike conventional convolutional neural networks that employ standard convolution operations throughout the network, MobileNetV2 introduces two important architectural innovations: depthwise separable convolution and inverted residual blocks with linear bottlenecks. These components substantially reduce the number of trainable parameters and floating-point operations while maintaining strong feature extraction capability. Depthwise separable convolution decomposes a standard convolution into two independent operations: depthwise convolution, which applies a separate convolutional filter to each input channel, and pointwise convolution, which employs a 1×1 convolution to combine information across channels. This decomposition significantly reduces computational complexity compared with conventional convolution. Mathematically, the computational cost of a standard convolution is expressed as

$$D_K^2 \times M \times N \times D_F^2 \quad (\text{Eq.1.})$$

where D_K denotes the kernel size, M represents the number of input channels, N denotes the number of output channels, and D_F is the spatial dimension of the feature map. By contrast, the computational cost of depthwise separable convolution is

$$D_K^2 \times M \times D_F^2 + M \times N \times D_F^2 \quad (\text{Eq.2.})$$

which substantially reduces the number of multiplication operations while preserving comparable feature extraction capability. Another key innovation is the inverted residual block, in which low-dimensional feature representations are first expanded into a higher-dimensional space, processed through depthwise convolution, and subsequently projected back into a compact representation using a linear bottleneck layer. This design enables efficient

information flow while reducing information loss during dimensionality reduction. Figure 3.5 illustrates the basic structure of the MobileNetV2 inverted residual block.

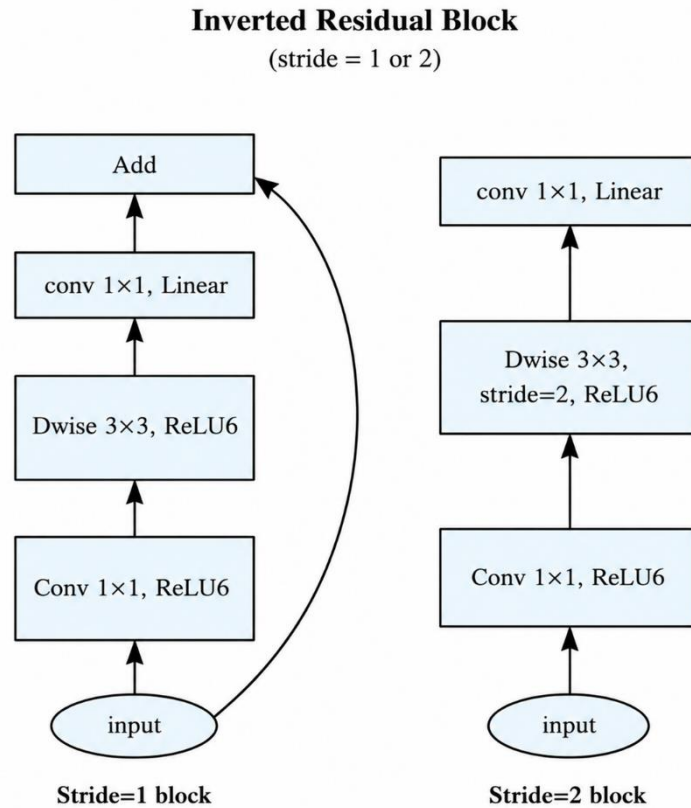


Figure 3.5: Architecture of MobileNetV2 Inverted Residual Block.

Within the proposed framework, MobileNetV2 functions as the shared feature extractor. The original fully connected classification layer was removed and replaced by the proposed multi-task prediction heads. Images resized to $224 \times 224 \times 3$ were fed into the pretrained backbone to generate shared feature representations for both food recognition and nutrition estimation. Training was conducted in two stages. During the first stage, the entire MobileNetV2 backbone remained frozen while only the task-specific prediction heads were optimized. In the second stage, the last 20 convolutional layers were unfrozen and jointly fine-tuned using a lower learning rate. This gradual adaptation enables the pretrained backbone to learn food-specific visual representations while preserving the generic image features acquired during ImageNet pretraining.

3.6.3. ResNet50

To investigate whether a deeper convolutional architecture improves multi-task learning performance, ResNet50 was selected as the second backbone network. Proposed by

He et al., ResNet50 [11] introduced residual learning, which enables very deep neural networks to be trained effectively by alleviating the vanishing gradient problem. Traditional deep convolutional networks often experience degradation in training accuracy as network depth increases. Rather than improving performance, simply stacking additional convolutional layers may make optimization increasingly difficult due to gradient attenuation during backpropagation. ResNet50 addresses this issue through identity shortcut connections, commonly referred to as skip connections. Instead of learning the desired mapping directly, each residual block learns a residual function.

$$F(x) = H(x) - x \quad (\text{Eq.3.})$$

where $H(x)$ represents the desired mapping and x denotes the block input. Consequently, the final output of each residual block becomes:

$$y = F(x) + x \quad (\text{Eq.4.})$$

Allowing gradients to propagate more effectively through the network during training. This residual learning mechanism facilitates optimization of substantially deeper networks while maintaining stable convergence. Figure 3.6 illustrates the architecture of a typical residual block.

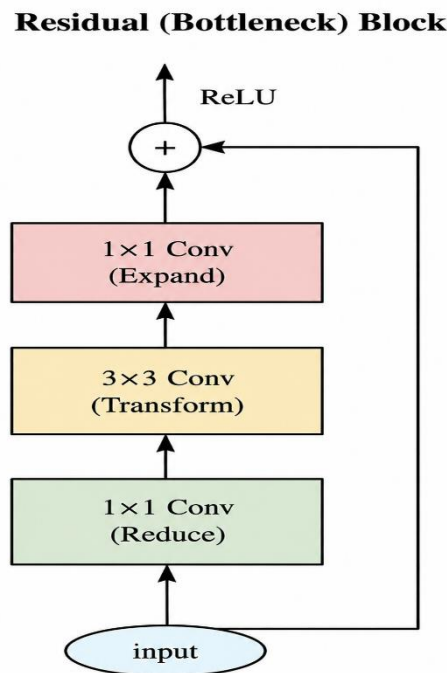


Figure 3.6: Residual Block in ResNet50

Within the proposed framework, the original fully connected classification layer of ResNet50 was removed, allowing the convolutional backbone to function as the shared feature extractor. Similar to MobileNetV2, the extracted visual features were simultaneously utilized by both the classification and regression branches. The same two-stage transfer learning strategy was adopted to ensure a fair comparison between the two backbone architectures. During the initial training stage, the ResNet50 backbone remained frozen while the task-specific prediction heads were optimized. Subsequently, the last 30 convolutional layers were unfrozen and jointly fine-tuned using a reduced learning rate. This strategy enables deeper convolutional layers to adapt to domain-specific food image characteristics while preserving robust generic visual representations learned from ImageNet. Although ResNet50 introduces greater computational complexity than MobileNetV2, its deeper architecture provides stronger feature representation capacity, which is expected to improve recognition of visually similar food categories and enhance overall classification performance as showed in Table 3.2.

Table 3.2: Comparison between MobileNetV2 and ResNet50

Characteristic	MobileNetV2	ResNet50
Publication	Sandler et al. (2018) [14]	He et al. (2016) [11]
Number of Layers	53	50
Main Innovation	Depthwise Separable Convolution, Inverted Residual	Residual Learning, Skip Connections
Parameters	~3.5 Million	~25.6 Million
Computational Cost	Low	High
Memory Requirement	Low	High
Suitable for	Mobile and Embedded Systems	High-performance Computing
Fine-tuned Layers in This Study	Last 20 Layers	Last 30 Layers

3.7. Model Training Strategy

Following data preprocessing and model construction, the proposed multi-task learning framework was trained to simultaneously perform multi-label food recognition and nutrition estimation. Model training aims to optimize the network parameters by minimizing a combined objective function while enabling the shared backbone to learn generalized visual representations that are beneficial to both prediction tasks. Training deep convolutional neural networks presents several challenges, particularly when the available training dataset is relatively limited. Training an entire backbone network from randomly initialized weights often leads to slow convergence, unstable optimization, and overfitting. Although transfer

learning provides an effective initialization through pretrained ImageNet weights, directly fine-tuning the entire network from the beginning may still cause catastrophic forgetting, whereby the pretrained visual representations are rapidly overwritten before the newly added prediction heads have learned meaningful task-specific mappings. To address these challenges, this study employed a two-stage transfer learning strategy, in which the prediction heads were first trained while the backbone remained frozen, followed by partial fine-tuning of the deeper convolutional layers. This strategy enables stable optimization while gradually adapting pretrained visual representations to the food image domain.

3.7.1. Two-Stage Fine-Tuning Strategy

Rather than training the entire network simultaneously, the proposed framework adopted a two-stage optimization strategy. This approach enables the newly introduced prediction heads to stabilize before modifying the pretrained backbone parameters.

- **Stage One - Frozen Backbone Training:** During the initial training stage, all convolutional layers of the pretrained backbone were frozen. Consequently, only the parameters within the classification and regression branches were updated during backpropagation. The purpose of this stage is to allow the task-specific prediction heads to learn appropriate mappings from the pretrained feature representations without disrupting the generalized visual knowledge already learned from ImageNet. This strategy offers three important advantages:
 - It significantly reduces the number of trainable parameters.
 - It accelerates convergence during the early training phase.
 - It prevents large gradient updates from degrading pretrained visual features.
- **Stage Two - Partial Fine-Tuning:** Once the prediction heads achieved stable convergence, selective fine-tuning of the backbone was performed. Rather than unfreezing the entire network, only the deeper convolutional layers were updated. Specifically, MobileNetV2 with last 20 convolutional layers and ResNet50 with last 30 convolutional layers were unfrozen and optimized jointly with the prediction heads using a lower learning rate. Fine-tuning only higher-level convolutional layers enables the backbone to adapt to food-specific visual characteristics while preserving generic image representations learned during ImageNet pretraining. Compared with full-network fine-tuning, partial fine-tuning substantially reduces overfitting and improves model stability, particularly when training data are limited.

3.7.2. Loss Function Optimization

As the proposed framework simultaneously performs two different learning tasks, each task requires an appropriate objective function that reflects its prediction characteristics. The classification branch predicts multiple food categories and therefore uses Weighted Binary Cross-Entropy (WBCE) loss. Since the food dataset exhibits class imbalance, class-specific weights are incorporated into the loss function to assign greater importance to minority food categories during optimization. The regression branch estimates continuous nutritional values and is optimized using Huber Loss. Compared with Mean Squared Error (MSE), Huber Loss combines sensitivity to small prediction errors with robustness against outliers, making it particularly suitable for nutritional data that may contain extreme values.

To jointly optimize both learning objectives, the proposed framework minimizes a weighted combination of the classification and regression losses. The combined loss function is defined as

$$L_{Total} = w_1 L_{WBCE} + w_2 L_{Reg} \quad (\text{Eq.5.})$$

$$L_{MobileNetV2} = 0.7 L_{WBCE} + 0.3 L_{Reg} \quad (\text{Eq.6.})$$

$$L_{ResNet50} = 0.3 L_{WBCE} + 0.7 L_{Reg} \quad (\text{Eq.7.})$$

The nutrient-specific weights were empirically selected during preliminary model development and held constant across both architectures where L_{WBCE} denotes the weighted binary cross-entropy loss used for multi-label food recognition, L_{Reg} denotes the weighted Huber regression loss, and w_1 and w_2 control the relative contribution of the classification and regression tasks, respectively. Preliminary experiments indicated that the two backbone architectures benefited from different task-level loss balances. MobileNetV2 achieved the most stable overall performance using $w_1 = 0.7$ and $w_2 = 0.3$, whereas ResNet50 achieved improved regression performance using $w_1 = 0.3$ and $w_2 = 0.7$. These backbone-specific configurations were therefore adopted in the final evaluation. Then define the regression loss:

$$L_{Reg} = \sum_{k=1}^5 \alpha_k L_{Huber, k} \quad (\text{Eq.8.})$$

where $L_{Huber, k}$ is the Huber loss for the k^{th} nutritional target and α_k is its nutrient-specific weight. The weights were fixed as follows: calories =0.30, fats =0.25, carbohydrates =0.20, protein =0.15, and sodium =0.10. These values sum to one and were kept identical for

MobileNetV2 and ResNet50 to ensure that the comparison of task-level weighting remained controlled.

3.7.3. Optimization Algorithm

Model optimization was performed using the Adam optimizer, which combines the advantages of adaptive learning rate estimation and momentum-based optimization. Adam has become one of the most widely adopted optimization algorithms in deep learning because of its fast convergence and stable performance across a wide range of computer vision applications. During training, Adam updates each parameter according to estimates of the first and second moments of the gradients, allowing different parameters to adaptively adjust their learning rates throughout optimization. The parameter update rule is expressed as

$$\theta_{t+1} = \theta_t - \alpha \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t + \epsilon}} \quad (\text{Eq.9.})$$

where,

- θ_t denotes model parameters,
- α is the learning rate,
- $\widehat{m}_t$ represents the bias-corrected first moment estimate,
- $\widehat{v}_t$ denotes the bias-corrected second moment estimate,
- ϵ is a small constant for numerical stability.

The Adam optimizer was selected because it provides efficient convergence while reducing sensitivity to learning rate selection compared with conventional stochastic gradient descent.

3.7.4. Hyperparameter Configuration

The selection of hyperparameters plays an important role in determining model convergence, prediction performance, and computational efficiency. Rather than employing arbitrary parameter values, hyperparameters were selected based on preliminary experiments and commonly adopted practices in transfer learning for computer vision as showed in Table 3.3.

Table 3.3: Hyperparameter Configuration for Model Training

Category	Hyperparameter	MobileNetV2	ResNet50
Input	Input image size	$224 \times 224 \times 3$	$224 \times 224 \times 3$
	Batch size	16	16
	Optimizer	Adam	Adam
	Initial learning rate	0.001	0.001

Category	Hyperparameter	MobileNetV2	ResNet50
Transfer Learning	Fine-tuning learning rate	0.0001	0.0001
	Stage 1	Freeze all backbone layers	Freeze all backbone layers
	Stage 2	Unfreeze last 20 layers	Unfreeze last 30 layers
	Activation function	Sigmoid	Sigmoid
Classification Head	Loss function	Weighted Binary Cross-Entropy	Weighted Binary Cross-Entropy
Regression Head	Number of output classes	42	42
	Activation function	Linear	Linear
	Loss function	Weighted Huber Loss	Weighted Huber Loss
	Nutrition targets	Calories, Fats, Carbohydrates, Protein, Sodium	Calories, Fats, Carbohydrates, Protein, Sodium
Task Loss Weighting	Classification loss weight (w_1)	0.70	0.30
	Regression loss weight (w_2)	0.30	0.70
	Calories	0.30	0.30
Regression Loss Weighting	Fats	0.25	0.25
	Carbohydrates	0.20	0.20
	Protein	0.15	0.15
	Sodium	0.10	0.10
Regularization	Batch Normalization	Yes	Yes
	Dropout	0.30, 0.20	0.30, 0.20
	Early Stopping	Patience = 5	Patience = 5
Training Strategy	ReduceLROnPlateau	Factor = 0.5, Patience = 2	Factor = 0.5, Patience = 2
	Model checkpoint	Best validation model	Best validation model

Table 3.3 summarizes the hyperparameter settings used in the proposed multi-task learning framework. Both backbone networks share the same input configuration, optimizer, learning rates, network heads, regression target weights, and regularization strategy. The primary differences lie in the number of unfrozen backbone layers during fine-tuning and the task-level loss weights. MobileNetV2 achieved its best overall performance using a classification-oriented loss balance ($w_1 = 0.7$, $w_2=0.3$), whereas ResNet50 achieved superior nutrition estimation performance using a regression-oriented loss balance ($w_1 = 0.3$, $w_2=0.7$). The nutrient-specific weights within the weighted Huber loss remained identical for both architectures to ensure a fair comparison.

3.7.5. Regularization Techniques

To improve model generalization and reduce overfitting, several regularization techniques were incorporated during training. First, dropout layers were applied within the fully connected prediction heads to randomly deactivate neurons during training. This prevents excessive co-adaptation among neurons and encourages the network to learn more robust feature representations. Second, batch normalization was employed after feature extraction layers to stabilize gradient propagation and accelerate convergence by reducing internal covariate shift. Third, early stopping monitored validation loss throughout training. When validation performance no longer improved for a predefined number of epochs, training was terminated automatically to prevent overfitting. Finally, ReduceLROnPlateau dynamically reduced the learning rate whenever validation performance stagnated, allowing the optimization process to converge more smoothly toward an optimal solution. Together, these regularization strategies contribute to improved model stability, enhanced generalization capability, and more reliable prediction performance on unseen food images.

3.8. Performance Evaluation

The performance of the proposed multi-task learning framework was evaluated using separate evaluation criteria for the food recognition and nutrition estimation tasks. Since the framework simultaneously performs multi-label classification and continuous-value regression, a single evaluation metric cannot adequately represent overall model performance. Consequently, different performance measures were adopted according to the prediction characteristics of each task. The food recognition component was formulated as a multi-label classification problem, in which each input image may contain multiple food categories simultaneously. Therefore, evaluation metrics commonly used in binary and multi-label classification were employed to assess the model's ability to correctly identify food components. These include Accuracy, Precision, Recall, F1-score, Hamming Accuracy, and Area Under the Receiver Operating Characteristic Curve (AUC). In contrast, nutrition estimation was formulated as a regression problem that predicts continuous nutritional values. Accordingly, regression performance was evaluated using Mean Absolute Error (MAE) and the Coefficient of Determination (R^2), which respectively measure prediction error and the proportion of variance explained by the model. The evaluation was conducted using the independent testing dataset, which was not utilized during model training or validation. This

testing strategy provides an unbiased assessment of the proposed framework's generalization capability on previously unseen food images.

3.8.1. Evaluation Strategy

To ensure a fair and objective assessment of model performance, the proposed framework was evaluated using three independent datasets: the training set, validation set, and testing set. During model development, the training dataset was used to optimize model parameters, while the validation dataset was employed to monitor learning progress, tune hyperparameters, and implement early stopping to prevent overfitting. Final performance evaluation was conducted exclusively on the testing dataset, which remained unseen throughout the training process. For the multi-label food recognition task, the sigmoid activation function produces an independent probability for each food category. A classification threshold of 0.5 was applied, whereby probabilities greater than or equal to 0.5 were classified as positive predictions and probabilities below 0.5 were classified as negative predictions. This threshold represents the conventional decision boundary in multi-label classification and provides a balance between false positive and false negative predictions. As food recognition and nutrition estimation represent fundamentally different learning tasks, their performance was evaluated independently using task-specific metrics. Classification metrics assess the model's ability to correctly identify food categories, whereas regression metrics evaluate the accuracy of predicted nutritional values. This dual evaluation strategy provides a comprehensive assessment of the proposed multi-task learning framework.

3.8.2. Evaluation of Multi-Label Food Recognition

The food recognition task was evaluated using six complementary performance metrics: Accuracy, Precision, Recall, F1-score, Hamming Accuracy, and Area Under the ROC Curve (AUC). These metrics provide different perspectives on model performance and collectively offer a comprehensive assessment of multi-label classification capability.

Table 3.4: Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

where,

- $TP = \text{True Positive}$
- $TN = \text{True Negative}$
- $FP = \text{False Positive}$
- $FN = \text{False Negative}$

- **Accuracy:** Accuracy measures the proportion of correctly classified predictions among all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (\text{Eq.10.})$$

where,

- TP represents True Positive
- TN represents True Negative
- FP represents False Positive
- FN represents False Negative

- **Precision:** Precision measures the proportion of predicted positive food labels that are actually correct. A high precision value indicates that the model generates relatively few false positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (\text{Eq.11.})$$

- **Recall:** Recall measures the proportion of actual food categories that are successfully detected by the model. Higher recall indicates stronger capability in identifying food categories that are present within an image.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (\text{Eq.12.})$$

- **F1-score:** The F1-score represents the harmonic mean of Precision and Recall. The F1-score is particularly appropriate for imbalanced datasets because it simultaneously considers both false positives and false negatives.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (\text{Eq.13.})$$

- **Hamming Accuracy:** Unlike conventional accuracy, Hamming Accuracy evaluates prediction correctness at the individual label level across all food categories. Hamming Accuracy is particularly suitable for multi-label classification because it evaluates each label independently rather than requiring all labels within an image to be predicted correctly.

$$\text{Hamming Accuracy} = 1 - \frac{1}{N \times L} \sum_{i=1}^N \sum_{j=1}^L I(y_{ij} \neq \hat{y}_{ij}) \quad (\text{Eq.14.})$$

where,

- N denotes the number of samples,
 - L represents the number of food categories,
 - y_{ij} is the ground truth label,
 - $\hat{y}_{ij}$ is the predicted label,
 - $I()$ represents the number of food categories.
- **Area Under the ROC Curve (AUC):** The Area Under the Receiver Operating Characteristic Curve (AUC) evaluates the model's ability to distinguish between positive and negative classes across all classification thresholds. An AUC value approaching 1.0 indicates excellent discriminative ability, whereas an AUC of 0.5 suggests performance equivalent to random guessing.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \quad (\text{Eq.15.})$$

where,

$$\text{TPR} = \frac{TP}{TP + FN} \quad (\text{Eq.16.})$$

and

$$\text{FPR} = \frac{FP}{FP + TN} \quad (\text{Eq.17.})$$

3.8.3. Evaluation of Nutrition Estimation

Unlike food recognition, nutrition estimation predicts continuous numerical values. Therefore, regression metrics were adopted to evaluate prediction accuracy.

- **Mean Absolute Error (MAE):** The Mean Absolute Error measures the average magnitude of prediction error. Lower MAE values indicate more accurate nutritional predictions.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (\text{Eq.18.})$$

- **Coefficient of Determination (R^2):** The coefficient of determination evaluates the proportion of variance explained by the regression model. An R^2 value close to 1 indicates excellent predictive performance, whereas values close to zero indicate limited explanatory power. Negative values suggest that the model performs worse than predicting the mean of the target variable.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (\text{Eq.19.})$$

3.8.4. Interpretation of Evaluation Metrics

Since the proposed framework performs two fundamentally different prediction tasks, no single evaluation metric is sufficient to fully characterize model performance. Classification metrics emphasize different aspects of food recognition, while regression metrics assess numerical prediction accuracy. For food recognition, Precision reflects the reliability of predicted food categories, Recall measures the ability to detect existing food items, and the F1-score balances these two objectives. Hamming Accuracy evaluates prediction correctness across individual labels, whereas AUC measures the model's overall discriminative capability independent of the classification threshold. For nutrition estimation, MAE quantifies the average prediction error in the original measurement units, making it straightforward to interpret practical prediction accuracy. Meanwhile, the coefficient of determination (R^2) evaluates how effectively the model explains variability in nutritional values across different food images. Using both metrics provides complementary insights into regression performance. Overall, the combination of classification and regression metrics enables a comprehensive and balanced evaluation of the proposed multi-task learning framework, ensuring that both food recognition performance and nutrition estimation accuracy are assessed objectively.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents and discusses the experimental results obtained from the proposed multi-task deep learning framework for simultaneous food recognition and nutrition estimation using 2D food images. The performance of two transfer learning architectures, namely MobileNetV2 and ResNet50, was systematically evaluated to investigate their effectiveness in addressing the two prediction tasks. The evaluation was conducted using the independent testing dataset described in Chapter 3. For the food recognition task, the performance of each model was assessed using multi-label classification metrics, including Accuracy, Precision, Recall, F1-score, Hamming Accuracy, and Area Under the Receiver Operating Characteristic Curve (AUC). For the nutrition estimation task, regression performance was evaluated using Mean Absolute Error (MAE) and the Coefficient of Determination (R^2). These complementary metrics provide a comprehensive assessment of both classification capability and nutritional prediction accuracy.

This chapter first presents the experimental setup and implementation environment, followed by a detailed evaluation of food recognition performance. Subsequently, the nutrition estimation results are analyzed and compared between the two backbone networks. Additional analyses, including per-class classification performance, regression error analysis, and comparisons with previous studies, are then discussed to provide a deeper understanding of the strengths and limitations of the proposed framework.

4.1. Experimental Setup

This section summarizes the experimental environment and implementation settings adopted to evaluate the proposed multi-task learning framework. The experiments were designed to provide a fair comparison between MobileNetV2 and ResNet50 by maintaining identical datasets, preprocessing procedures, training strategies, and evaluation metrics. Consequently, any differences in performance can be attributed primarily to the characteristics of the backbone architectures rather than differences in the experimental configuration. The proposed framework was implemented using the Python programming language together with the TensorFlow and Keras deep learning libraries. Model training and evaluation were performed using Google Colaboratory, which provides cloud-based GPU acceleration for computationally intensive deep learning tasks. Additional Python libraries, including NumPy,

Pandas, Matplotlib, and Scikit-learn, were utilized for data processing, visualization, and performance evaluation.

The experiments employed the NutritionVerse-Real dataset after completing the preprocessing procedures described in Chapter 3. The dataset consisted of cleaned food images together with corresponding dish-level nutritional information and multi-label food annotations. To ensure a consistent evaluation protocol, the dataset was divided into independent training, validation, and testing subsets. The training dataset was used to optimize model parameters, the validation dataset was employed for hyperparameter tuning and early stopping, and the testing dataset was reserved exclusively for final performance evaluation. This separation prevents information leakage and enables an unbiased assessment of the model's generalization capability. To ensure a fair comparison between the two backbone networks, both MobileNetV2 and ResNet50 were trained using the same experimental settings. Images were resized to 224×224 pixels, and the proposed two-stage transfer learning strategy was applied to both models. During the first stage, the pretrained backbone remained frozen while only the classification and regression heads were optimized. In the second stage, selective fine-tuning was performed by unfreezing the last 20 convolutional layers of MobileNetV2 and the last 30 convolutional layers of ResNet50. This strategy enables gradual adaptation of the pretrained feature representations to the food image domain while reducing the risk of overfitting.

Both architectures were optimized using the Adam optimizer with an initial learning rate of 0.001 during the first training stage and a reduced learning rate of 0.0001 during fine-tuning. The classification branch was trained using Weighted Binary Cross-Entropy (WBCE) loss to mitigate the effects of class imbalance, whereas the nutrition estimation branch employed Huber Loss to provide robustness against outliers in nutritional values. The total optimization objective was obtained by combining the two losses using the weighting strategy presented in Chapter 3. To further improve model generalization, several regularization techniques were incorporated during training. These include data augmentation (for the augmented experiments), dropout, batch normalization, early stopping, and adaptive learning rate reduction using the ReduceLROnPlateau callback. The same regularization strategy was applied to both backbone architectures to ensure consistency throughout the comparative analysis. With the experimental environment and implementation settings established, the following sections present a comprehensive evaluation of the proposed multi-task learning framework. The analysis begins with the food recognition performance of MobileNetV2 and

ResNet50, followed by a detailed comparison of their nutrition estimation capabilities and an in-depth discussion of the observed results.

4.2. Food Recognition Performance

Food recognition constitutes the first stage of the proposed multi-task learning framework and serves as the foundation for subsequent nutrition estimation. Since a single dish may contain multiple food components, the task was formulated as a multi-label classification problem, requiring the model to identify all food categories present within an image rather than assigning only one label. Accurate food recognition is essential because the predicted food labels are subsequently incorporated into the regression branch to assist nutritional value estimation. This section evaluates the food recognition performance of the proposed framework using two transfer learning backbone networks, namely MobileNetV2 and ResNet50. The evaluation was conducted on the independent testing dataset using Hamming Accuracy, Micro Precision, Micro Recall, Micro F1-score, Macro F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC). These metrics provide complementary perspectives on overall prediction accuracy, class-level performance, and the model's discrimination capability in handling multi-label classification problems with imbalanced data. The following subsections present the overall classification performance of both backbone networks, followed by analyses of the learning behaviour during training, receiver operating characteristic (ROC) curves, and per-class classification performance. The findings are subsequently discussed to provide a comprehensive understanding of the strengths and limitations of the proposed food recognition framework.

4.2.1. Overall Classification Results

The overall food recognition performance of MobileNetV2 and ResNet50, with data augmentation applied during training, is presented in Table 4.1. The evaluation was conducted on the independent testing dataset to ensure that the reported results reflect the models' generalization capability rather than their ability to memorize the training data.

Table 4.1: Food Recognition Performance of the Proposed Models

Metric	MobileNetV2	ResNet50
Hamming Accuracy	0.9132	0.9071
Micro Precision	0.5034	0.4715
Micro Recall	0.4545	0.5273
Micro F1-Score	0.4777	0.4979
Macro F1-Score	0.3414	0.3905
Micro AUC	0.8938	0.8929

Table 4.1 presents the overall multi-label food recognition performance of MobileNetV2 and ResNet50 on the independent testing set. Both architectures achieved comparable Hamming Accuracy and Micro AUC values, indicating strong label-level correctness and discrimination capability. MobileNetV2 obtained a Hamming Accuracy of 0.9132, Micro Precision of 0.5034, and Micro AUC of 0.8938, while ResNet50 achieved corresponding values of 0.9071, 0.4715, and 0.8929. Although MobileNetV2 produced slightly higher precision, Hamming Accuracy, and AUC, ResNet50 achieved substantially higher Micro Recall of 0.5273. This improvement resulted in higher Micro F1 and Macro F1 scores of 0.4979 and 0.3905, compared with 0.4777 and 0.3414 for MobileNetV2. The higher recall indicates that ResNet50 identified a greater proportion of the food components actually present in the test images, although this broader prediction behavior also introduced slightly more false-positive labels and reduced precision. For this imbalanced multi-label task, the higher Micro and Macro F1 scores indicate that ResNet50 achieved the stronger overall food recognition performance. In particular, the improvement in Macro F1 suggests better balance across food categories, including less frequently represented classes. Nevertheless, MobileNetV2 remained competitive and demonstrated a more conservative prediction behavior, together with substantially lower computational complexity. Overall, ResNet50 achieved stronger balanced food recognition performance, whereas MobileNetV2 maintained slightly higher precision, Hamming Accuracy, and AUC with considerably lower computational cost.

To isolate the effect of data augmentation from architecture and loss-weighting choices, both backbones were retrained using only the original, non-augmented training images (711 images), with all other settings including each backbone's tuned loss weights and unfreeze depth held fixed. Table 4.2 compares classification performance with and without augmentation for both backbones.

Table 4.2: Classification Performance With vs. Without Data Augmentation

Metric	Model	No Augmentation	Augmentation	Change
Hamming Accuracy	MobileNetV2	0.9111	0.9132	0.0021
	ResNet50	0.8802	0.9071	0.0269
Micro Precision	MobileNetV2	0.4915	0.5034	0.0119
	ResNet50	0.3427	0.4715	0.1288
Micro Recall	MobileNetV2	0.5242	0.4545	-0.0697
	ResNet50	0.4061	0.5273	0.1212
Micro F1-Score	MobileNetV2	0.5073	0.4777	-0.0296
	ResNet50	0.3717	0.4979	0.1262
Macro F1-Score	MobileNetV2	0.3905	0.3414	-0.0491
	ResNet50	0.2817	0.3905	0.1088

The effect of augmentation on classification is the reverse of its effect described for regression in Section 4.3.1: here, augmentation benefits ResNet50 far more than MobileNetV2. Before augmentation reduced every ResNet50 classification metric substantially Micro F1 fell from 0.4979 to 0.3717 and Macro F1 from 0.3905 to 0.2817 consistent with ResNet50's larger capacity (25.6M parameters) overfitting the smaller, non-augmented training set. For MobileNetV2, by contrast, before augmentation produced a small, consistent improvement across every metric: Micro F1 increased from 0.4777 to 0.5073 and Macro F1 from 0.3414 to 0.3905. Despite this classification performance loss for MobileNetV2, augmentation is retained in the training pipeline because the regression improvements substantially outweigh the classification decline. For MobileNetV2, augmentation improved regression head since nutrition estimation is the primary end-goal of this multi-task framework, this trade-off is justified: a $\sim 2\%$ reduction in food recognition metrics is acceptable in exchange for meaningfully improved nutritional predictions, which directly impact dietary guidance and health outcomes. This pragmatic prioritization reflects the application context users deploying such systems care more about accurate nutrition estimation than perfect food classification. Taken together with the regression results in Section 4.3.1, augmentation is unambiguously beneficial for ResNet50 across both tasks, whereas for MobileNetV2 it represents a deliberate trade-off: beneficial for the primary task (regression head) but not for the secondary task (classification head). This asymmetry reinforces the broader pattern observed throughout this study: the higher-capacity ResNet50 backbone consistently requires more support whether through loss-weight tuning or data augmentation to perform reliably on this dataset, while MobileNetV2 is comparatively robust without it. For practitioners deploying similar multi-task food analysis systems, this finding suggests that augmentation strategy should be determined by task priority rather than uniform application across all objectives.

4.2.2. Training and Validation Curve Analysis

The learning curves provide further insight into the optimization behavior and generalization capability of the proposed multi-task learning framework throughout the two-stage transfer learning process. Figures 4.1 to 4.4 illustrate the training and validation performance of MobileNetV2 and ResNet50 using three complementary indicators, namely the overall loss, classification AUC, and nutrition estimation MAE. The vertical dashed line in each figure indicates the transition from the frozen-backbone training stage to the fine-tuning

stage, where the final convolutional layers of the pretrained backbone were unfrozen for domain adaptation.

- **Loss Curve Analysis**

As illustrated in Figure 4.1, the training and validation losses of MobileNetV2 decreased steadily throughout the frozen training stage, indicating that the model successfully learned meaningful feature representations from the food images. After fine-tuning commenced, the training loss briefly increased before continuing to decrease, while the validation loss remained relatively stable with only a slight increase during the final epochs. This behavior is expected when additional pretrained layers become trainable and suggests that the model adapted successfully to the target domain without exhibiting severe overfitting.

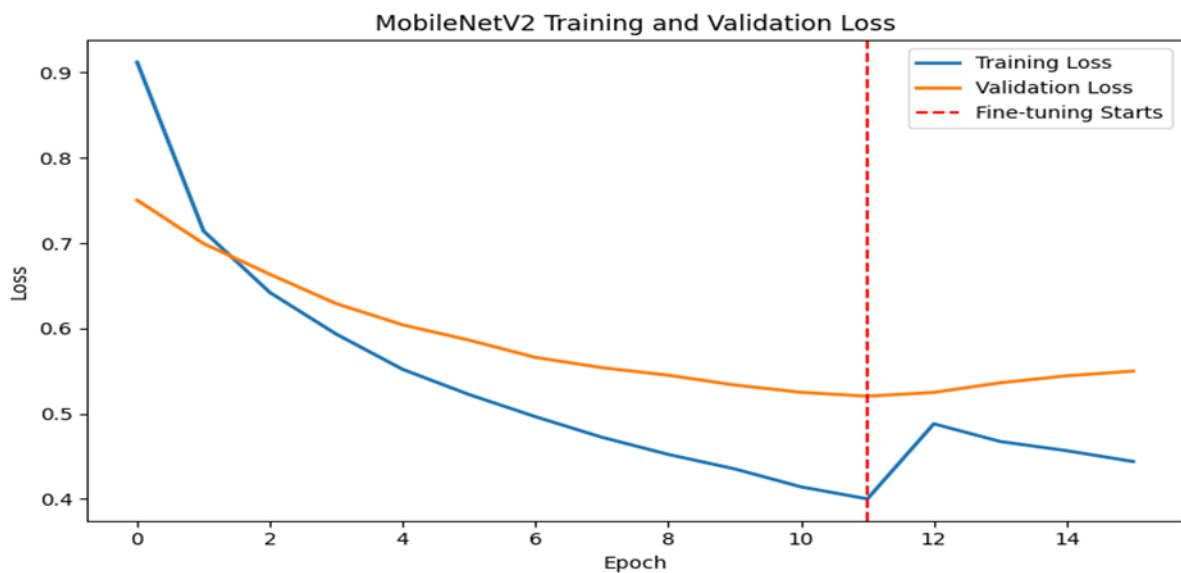


Figure 4.1: Training and Validation Loss MobileNetV2

A similar convergence pattern was observed for ResNet50, as shown in Figure 4.2. Both training and validation losses consistently decreased during the initial training stage and remained stable after fine-tuning. Compared with MobileNetV2, the validation loss of ResNet50 exhibited smaller fluctuations during the later epochs, indicating a more stable optimization process. The deeper residual architecture appears to facilitate more consistent feature refinement while maintaining good generalization throughout training.

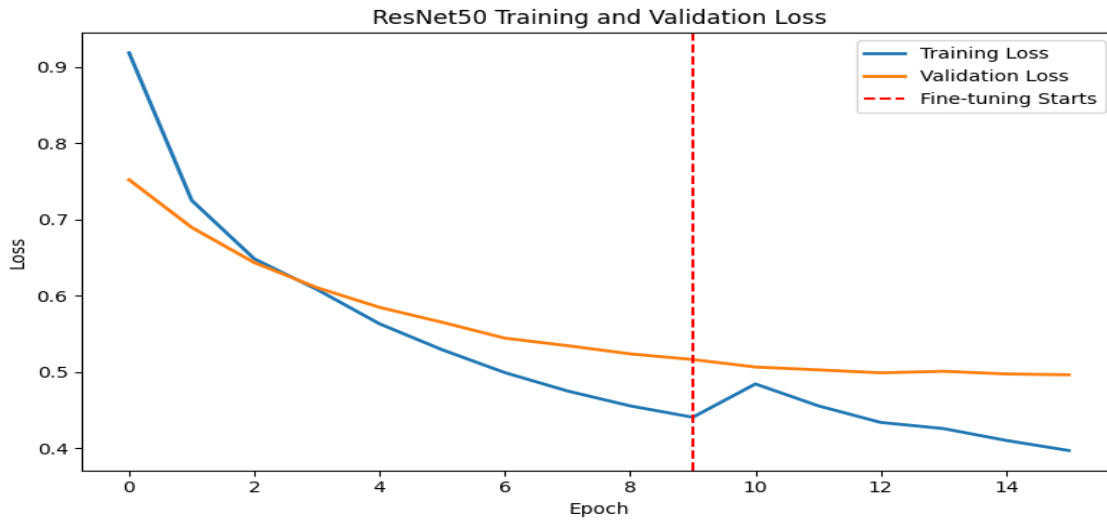


Figure 4.2: Training and Validation Loss of ResNet50

- **Classification AUC Analysis**

Figure 4.3 shows the classification AUC obtained by MobileNetV2 during training. Both the training and validation AUC increased steadily throughout the optimization process before gradually converging after fine-tuning. Although the training AUC continued to improve, the validation AUC stabilized near the final epochs, suggesting that additional training produced only marginal improvements in generalization. Nevertheless, the consistently increasing validation AUC demonstrates that the model maintained strong discrimination capability for the multi-label food recognition task.

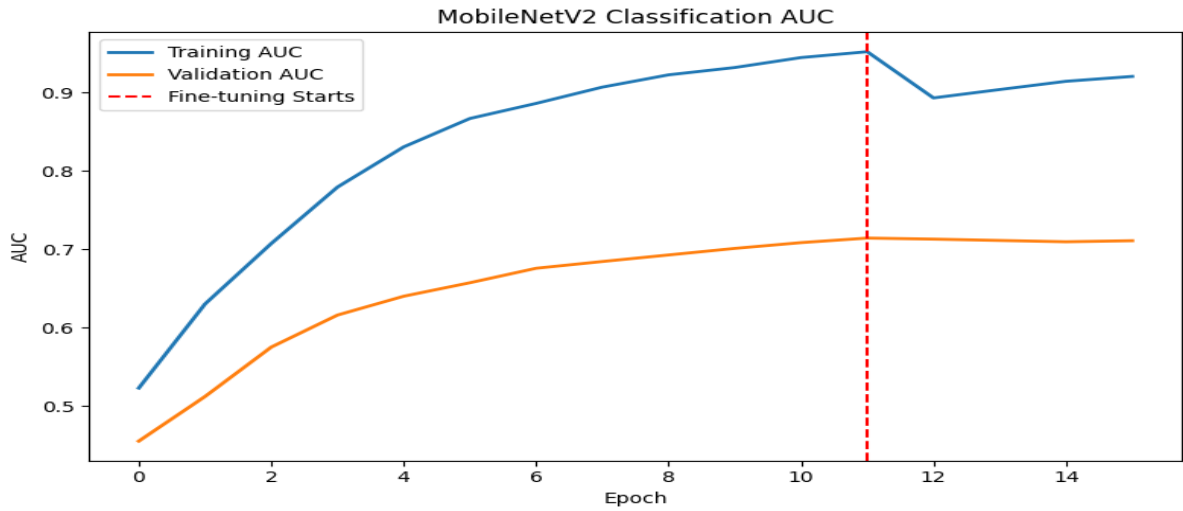


Figure 4.3: Classification AUC of MobileNetV2

As presented in Figure 4.4, ResNet50 exhibited a similar learning trend but achieved a slightly higher validation AUC than MobileNetV2. The validation AUC increased steadily across both training stages before converging at approximately 0.74, while the training AUC approached 0.97 during the final epochs. The smooth convergence of the validation curve indicates that the deeper residual architecture effectively learned discriminative visual representations without introducing substantial instability during fine-tuning.

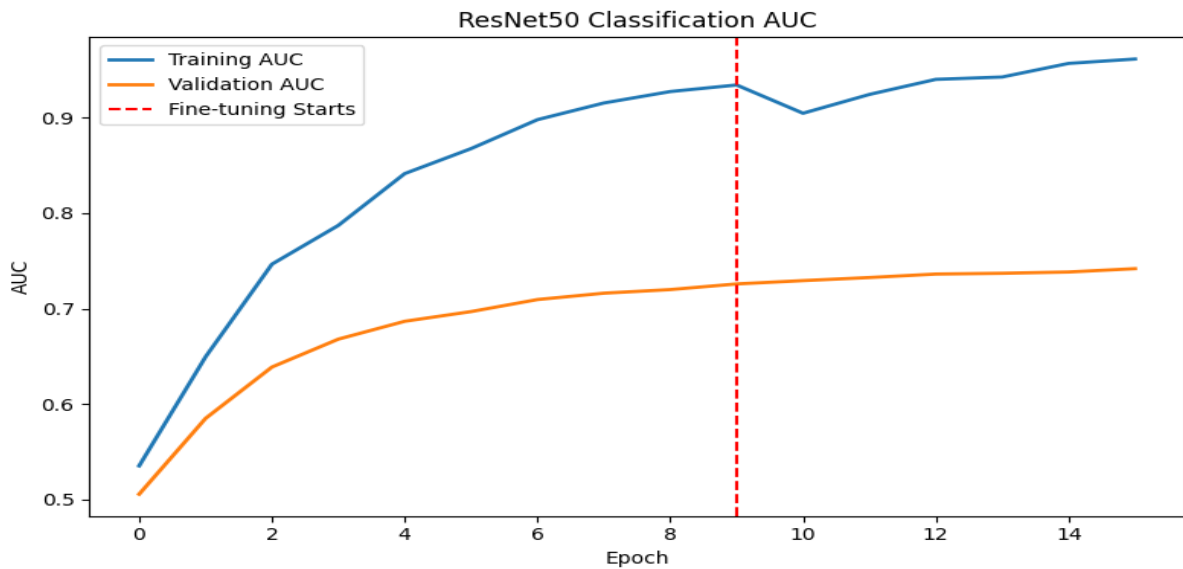


Figure 4.4: Classification AUC of ResNet50

- **Overall Discussion**

Overall, the learning curves demonstrate that both MobileNetV2 and ResNet50 converged successfully under the proposed two-stage transfer learning strategy. The continuous reduction in the training and validation losses, together with the progressive improvement in the classification AUC, indicates that both backbone networks effectively learned discriminative feature representations for the multi-label food recognition task. Minor fluctuations were observed immediately after the fine-tuning stage commenced, which are expected when previously frozen convolutional layers are unfrozen for domain adaptation and adapt to the target dataset. These temporary variations did not indicate optimization instability, as both models subsequently resumed stable convergence throughout the remaining training process. Although MobileNetV2 and ResNet50 exhibited slightly different convergence behaviors, both architectures demonstrated stable learning without clear evidence of severe overfitting. MobileNetV2 achieved stable optimization under the classification-oriented task weighting ($w_1 = 0.7$, $w_2 = 0.3$) whereas ResNet50 converge successfully using the regression-oriented configuration ($w_1 = 0.3$, $w_2 = 0.7$). The stable validation trends indicate that increasing the contribution of the regression task did not compromise optimization stability while enabling more effective learning of nutrition estimation features. These observations suggest that different backbone architectures may benefit from different task-loss weight configurations to achieve an appropriate balance between food recognition and nutrition estimation.

The consistent convergence observed throughout the training process further validates the effectiveness of the proposed two-stage transfer learning strategy presented in Chapter 3. Furthermore, the learning curves demonstrate that combining backbone-specific task-loss weighting with transfer learning, early stopping, adaptive learning rate scheduling, and regularization techniques enables both architectures to achieve stable optimization and good generalization on unseen food images. Having established the optimization behavior of both models during training, the following section evaluates their nutrition estimation performance using the independent testing dataset.

4.2.3. Receiver Operating Characteristic (ROC) Curve Analysis

The Receiver Operating Characteristic (ROC) curve provides a comprehensive evaluation of the classification capability of the proposed framework by illustrating the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) over a range of classification thresholds. Unlike evaluation metrics computed at a single decision threshold,

the ROC curve assesses the model's discrimination ability independently of a fixed classification boundary. For multi-label classification problems, the micro-average ROC curve aggregates predictions across all food categories, providing an overall measure of classification performance.

Figures 4.5 compares the ROC curves of MobileNetV2 and ResNet50 for multi-label food recognition. MobileNetV2 achieved a Micro AUC of 0.8938, while ResNet50 achieved a comparable Micro AUC of 0.8929. Both values are substantially higher than the random-classification baseline, indicating that both models effectively distinguish between the presence and absence of food labels across different classification thresholds. The difference in Micro AUC between the two models is minimal, suggesting that both backbone architectures possess similar overall discrimination capability. Although ResNet50 achieved higher micro recall and F1-score at the selected classification threshold, MobileNetV2 obtained a slightly higher Micro AUC and Micro Precision. This indicates that the two models exhibit different prediction characteristics rather than a clear superiority of one model over the other. Overall, the ROC analysis confirms that both architectures provide strong and comparable classification performance within the proposed multi-task learning framework.

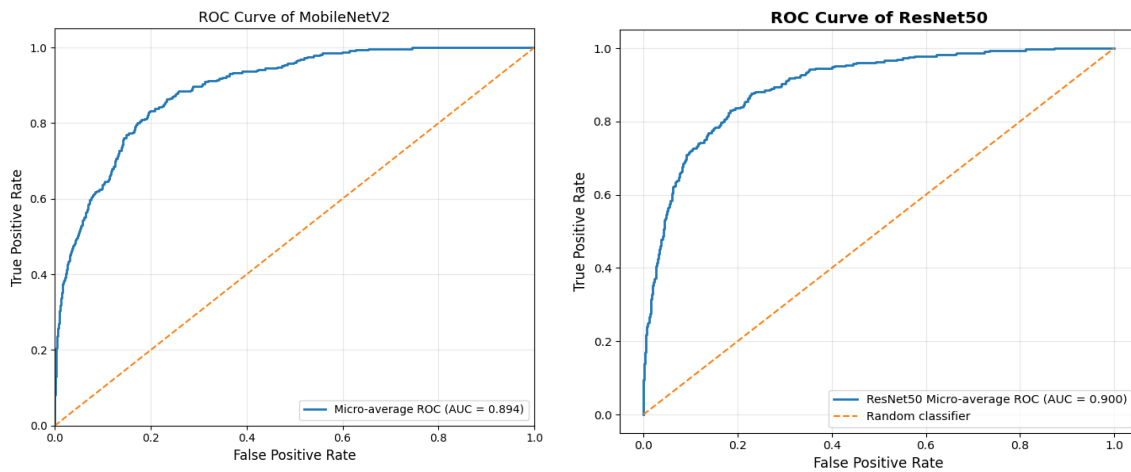


Figure 4.5: ROC Curve of MobileNetV2 and ResNet50

4.2.4. Per-Class Classification Analysis

Although the overall evaluation metrics demonstrate strong classification performance, they do not provide detailed information regarding the recognition performance of individual food categories. Therefore, a per-class analysis was conducted using the F1-score, which balances Precision and Recall and is particularly suitable for evaluating imbalanced multi-label

classification problems. This analysis enables a more detailed assessment of the strengths and limitations of the proposed framework across different food classes.

- **MobileNetV2 Per-Class Performance**

Figure 4.6 presents the per-class F1-score distribution of MobileNetV2. The results show that MobileNetV2 performed well on visually distinctive food categories, particularly those with clear shapes, textures, or colour patterns.

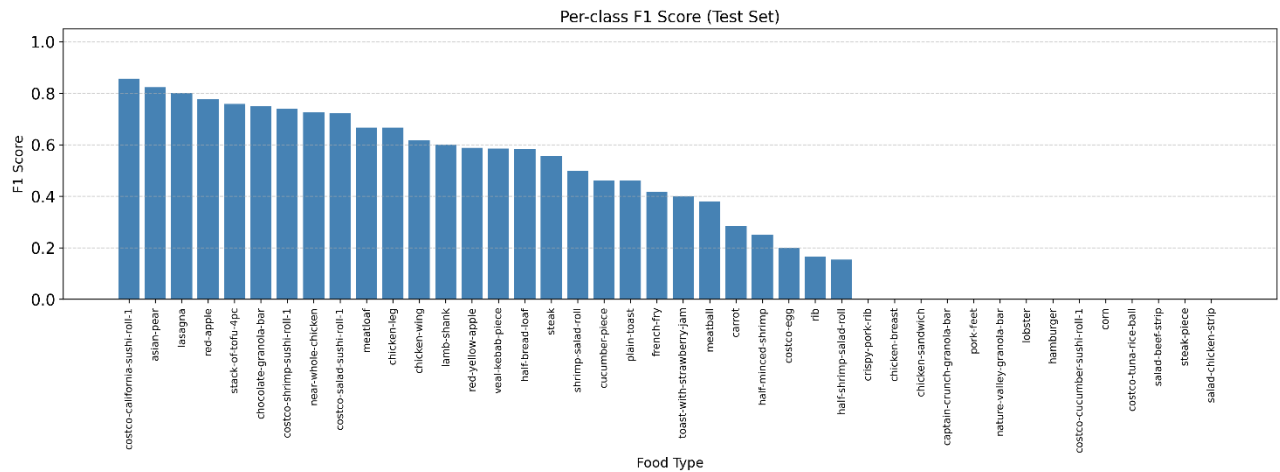


Figure 4.6: Per-Class F1-score Distribution of MobileNetV2

Table 4.3 : Top-10 and Bottom-10 Performing Food Classes for MobileNetV2

Top 10 Food Class	F1 Score	Bottom 10 Food Class	F1 Score
Lasagna	1.00	Captain-crunch-granola-bar	0.00
Shrimp-salad-roll	0.89	Carrot	0.00
Near-whole-chicken	0.86	Chicken-breast	0.00
Meatloaf	0.80	Corn	0.00
Asian-pear	0.74	Crispy-pork-rib	0.00
Costco-salad-sushi-roll-1	0.70	Pork-feet	0.00
Stack-of-tofu-4pc	0.69	Hamburger	0.00
Red-apple	0.67	Lobster	0.00
Cucumber-piece	0.67	Nature-valley-granola-bar	0.00
Half-bread-loaf	0.67	Salad-beef-strip	0.00

As shown in Table 4.3, the highest-performing classes included Lasagna, Shrimp-salad-roll, Near-whole-chicken, and Meatloaf, indicating that the model was able to learn strong visual representations for food categories with recognizable appearance patterns. However, there are several classes that achieved F1-scores of zero, including Captain-crunch-granola-bar, Carrot, Chicken-breast, Corn, Crispy-pork-rib, Pork-feet, Hamburger, Lobster, Nature-valley-granola-bar, and Salad-beef-strip. These low-performing classes were likely affected by

limited sample availability, class imbalance, and visual similarity with other food categories. This suggests that although MobileNetV2 provides efficient classification performance, it remains sensitive to minority classes and visually ambiguous food items.

- **ResNet50 Per-Class Performance**

Figure 4.7 illustrates the per-class F1-score distribution of ResNet50. Compared with MobileNetV2, ResNet50 achieved stronger performance across several food categories, reflecting its deeper residual architecture and greater feature extraction capacity.

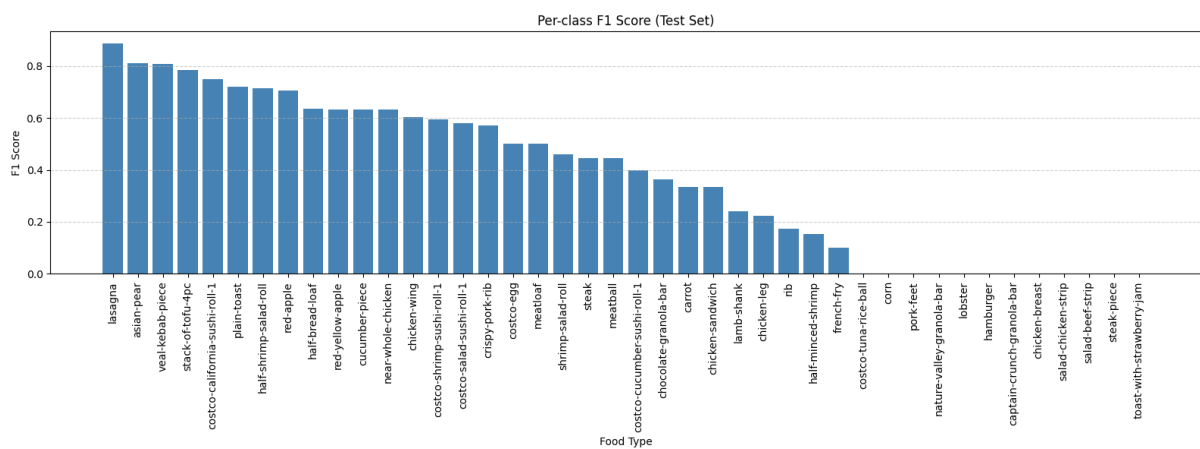


Figure 4.7: Per-Class F1-score Distribution of ResNet50

Table 4.4: Top-10 and Bottom-10 Performing Food Classes for ResNet50

Top 10 Food Class	F1 Score	Bottom 10 Food Class	F1 Score
Costco-california-sushi-roll-1	1.00	Chicken-breast	0.00
Near-whole-chicken	0.91	Carrot	0.00
Asian-pear	0.87	Corn	0.00
Lasagna	0.74	Hamburger	0.00
Plain-toast	0.74	Lobster	0.00
Meatloaf	0.73	Nature-valley-granola-bar	0.00
Veal-kebab-piece	0.73	Pork-feet	0.00
Half-bread-loaf	0.67	Salad-chicken-strip	0.00
Red-apple	0.67	Toast-with-strawberry-jam	0.00
Costco-cucumber-sushi-roll-1	0.67	Steak-piece	0.00

As shown in Table 4.4, the best-performing classes included Costco-california-sushi-roll-1, Near-whole-chicken, Asian-pear, Lasagna, Plain-toast, and Meatloaf. These results indicate that ResNet50 was able to capture more complex visual patterns and improve

recognition for several visually distinctive classes. Nevertheless, there are some classes that ResNet50 also struggled with some food categories, including Chicken-breast, Carrot, Corn, Hamburger, Lobster, Nature-valley-granola-bar, Pork-feet, Salad-chicken-strip, Toast-with-strawberry-jam, and Steak-piece. The persistence of low F1-scores for these classes indicates that increasing model depth alone does not fully solve the challenges caused by class imbalance and visual similarity.

- **Comparative Discussion**

Overall, the per-class analysis demonstrates that both models performed better on food categories with distinctive visual characteristics and struggled with minority or visually similar classes. ResNet50 generally produced stronger and more stable per-class performance than MobileNetV2, which is consistent with its higher overall Micro F1-score and Macro F1-score reported in Section 4.2.1. This improvement can be attributed to the deeper residual architecture, which enables richer hierarchical feature extraction. However, both models continued to produce near-zero F1-scores for several underrepresented classes. This finding indicates that the main limitation is not only model architecture but also dataset characteristics, particularly class imbalance, limited sample size, and visual ambiguity among certain food categories. Therefore, future improvements should focus on expanding minority class samples, improving data augmentation strategies, and collecting more diverse food images to enhance recognition performance across all classes.

Overall, the per-class classification analysis confirms that the proposed framework can recognize many food categories effectively, especially visually distinctive foods. However, the results also highlight that reliable multi-label food recognition remains challenging for underrepresented and visually similar food classes. This observation is important because classification outputs are used as conditioning information for the nutrition estimation branch; therefore, weaknesses in food recognition may influence the accuracy of subsequent nutritional prediction.

4.3. Nutrition Estimation Performance

While the previous section demonstrated that the proposed multi-task learning framework achieved satisfactory performance for multi-label food recognition, accurately estimating nutritional values from a single RGB image remains considerably more challenging. Unlike food recognition, which focuses on identifying the presence of food categories, nutrition

estimation requires the model to infer quantitative information, including calories, fats, carbohydrates, protein, and sodium, from visual appearance alone. These nutritional quantities are influenced not only by food identity but also by portion size, ingredient composition, cooking methods, and serving quantity, many of which cannot be directly observed from a two-dimensional image. In the proposed framework, the regression branch receives both the shared visual features and the predicted food classification outputs, enabling nutritional estimation to be conditioned on the recognized food components. This architectural design aims to exploit the relationship between food identity and nutritional content. Nevertheless, the absence of explicit depth and volume information presents a fundamental limitation for image-based nutrition estimation. Therefore, this section evaluates the regression performance of the proposed framework using MAE, RMSE, and the coefficient of determination (R^2), followed by analyses of the regression learning behavior and prediction accuracy across the five nutritional targets.

4.3.1. Overall Nutrition Estimation Results

The overall regression performance of MobileNetV2 and ResNet50 for the five nutritional targets is presented in Table 4.4. The evaluation was conducted using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), where lower MAE and RMSE values indicate smaller prediction errors, while higher R^2 values indicate better agreement between predicted and actual nutritional values.

Table 4.5: Overall Nutrition Estimation Performance

Nutrient	MobileNetV2			ResNet50		
	RMSE	MAE	R^2	RMSE	MAE	R^2
Calory (kCal)	1034.41	634.30	0.173	1038.69	631.36	0.166
Fats (g)	60.29	35.58	0.190	65.34	35.31	0.049
Carbs (g)	97.36	49.78	0.279	96.65	49.01	0.290
Protein (g)	113.31	59.58	0.054	112.20	54.68	0.072
Sodium (mg)	0.50	0.38	0.613	0.45	0.35	0.691

Table 4.4 presents the nutrition estimation performance of MobileNetV2 and ResNet50 across the five nutritional targets. Both architectures achieved positive R^2 values for every target, demonstrating that each model learned predictive relationships beyond the target-mean baseline. For MobileNetV2, sodium achieved the highest prediction performance with an R^2

of 0.613, followed by carbohydrates at 0.279, fats at 0.190, calories at 0.173, and protein at 0.054. ResNet50 similarly achieved its strongest performance for sodium, reaching an R^2 of 0.691, followed by carbohydrates at 0.291, calories at 0.166, protein at 0.072, and fats at 0.049. The comparison reveals complementary strengths between the two architectures. MobileNetV2 performed better for calories and fats, whereas ResNet50 achieved higher R^2 values for carbohydrates, protein, and sodium. ResNet50 also obtained lower MAE values for all five targets, although the differences were relatively small for several nutrients. The strong improvement compared with the earlier classification-focused ResNet50 configuration indicates that assigning greater weight to the regression objective allowed the shared backbone to learn representations that were more useful for quantitative nutrition estimation. Sodium and carbohydrates were the most predictable targets for both models, whereas protein remained difficult for MobileNetV2 and fats remained difficult for ResNet50. Calories also achieved only modest R^2 values in both architectures.

These results confirm that food identity provides useful nutritional information, but the absence of explicit portion-size and volume cues continues to limit the prediction of quantity-dependent nutrients from a single two-dimensional image. Overall, neither backbone achieved uniform superiority across all nutritional targets. MobileNetV2 provided stronger predictions for calories and fats, while the regression-weighted ResNet50 configuration produced stronger results for carbohydrates, protein, and sodium and achieved the highest individual R^2 value of 0.691 for sodium.

The same non-augmented configuration described in Section 4.2.1 was also evaluated for nutrition estimation, isolating the effect of augmentation on regression performance for both backbones. Table 4.6 reports R^2 and MAE with and without augmentation for each nutrient.

Table 4.6: Nutrition Estimation Performance With vs. Without Data Augmentation

Nutrient	Model	MAE Aug	MAE No Aug	R^2 Aug	R^2 No Aug	ΔR^2
Calories (kCal)	MobileNetV2	634.30	650.38	0.173	0.076	+0.097
	ResNet50	631.36	659.61	0.166	-0.005	+0.171
Fats (g)	MobileNetV2	35.58	35.69	0.190	-0.016	+0.206
	ResNet50	35.31	35.78	0.049	-0.054	+0.103
Carbs (g)	MobileNetV2	49.78	42.82	0.279	0.454	-0.175
	ResNet50	49.01	51.66	0.290	0.175	+0.115
Protein (g)	MobileNetV2	59.58	57.17	0.054	-0.061	+0.115
	ResNet50	54.68	63.47	0.072	-0.112	+0.184
Sodium (mg)	MobileNetV2	0.38	0.42	0.613	0.586	+0.027
	ResNet50	0.35	0.41	0.691	0.579	+0.112

The effect of augmentation differs sharply by backbone. For ResNet50, augmentation improved every nutrient without exception, and was decisive rather than marginal: R^2 for calories rose from -0.005 to 0.166 and for protein from -0.112 to 0.072, moving both from below-baseline to positive predictive value. This is consistent with ResNet50's larger capacity (25.6M parameters) making it more prone to overfitting the smaller, non-augmented training set, and indicates that augmentation and regression-dominant loss weighting (Section 3.7.2) address complementary aspects of the same underlying capacity or data-size imbalance. For MobileNetV2, the effect was mixed: augmentation improved calories, fats, protein, and sodium, but carbohydrates was better predicted without augmentation ($R^2 = 0.454$ versus 0.279 with augmentation). This suggests that, for a lower-capacity backbone already less prone to overfitting, augmentation trades a small amount of accuracy on some nutrients for improved stability on others rather than delivering uniform gains. Overall, augmentation was most clearly beneficial for the higher-capacity ResNet50 backbone, for which it was necessary to avoid negative R^2 values, whereas its benefit for MobileNetV2 was smaller and nutrient-dependent.

4.3.2. Learning Behavior of the Regression Branch

The learning behavior of the regression branch was further examined using the training and validation Mean Absolute Error (MAE) curves presented in Figures 4.8 and 4.9. These learning curves provide insight into the optimization process and the ability of the regression branch to generalize beyond the training data.

Figure 4.8 illustrates the Mean Absolute Error (MAE) of the nutrition estimation branch during training. The training and validation MAE decreased progressively during the frozen training stage, indicating continuous improvement in regression performance. Following fine-tuning, a temporary increase in training MAE was observed before the model stabilized, whereas the validation MAE remained relatively constant. This behavior suggests that the regression branch successfully adapted to the updated backbone features while preserving its generalization capability.

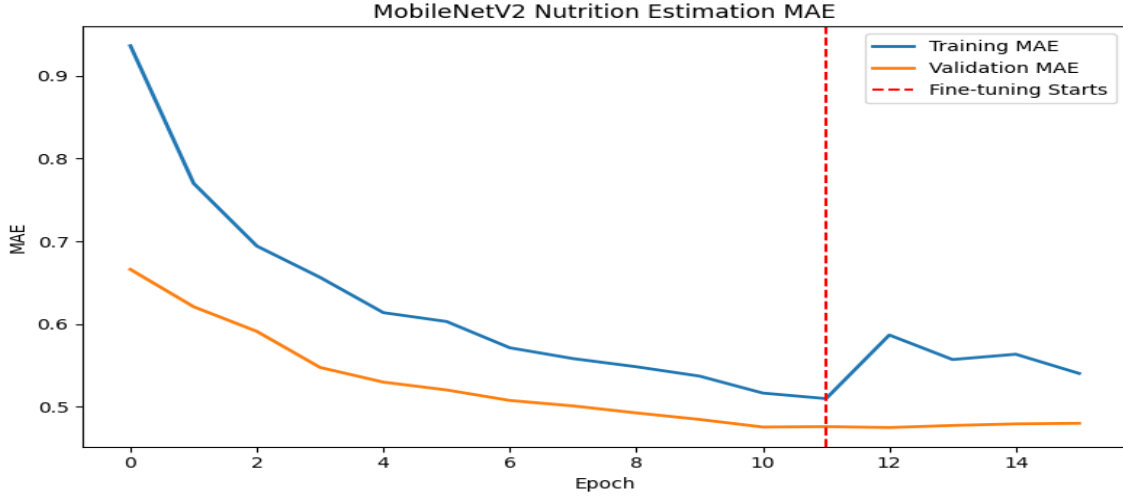


Figure 4.8: Nutrition Estimation MAE of MobileNetV2

Figure 4.9 illustrates the learning behavior of the ResNet50 regression branch under the revised task-loss configuration of $w_1 = 0.3$ and $w_2 = 0.7$. The training and validation MAE decreased throughout the optimization process, with limited fluctuations following the transition to partial fine-tuning. This behavior indicates that increasing the regression task weight provided a stronger optimization signal to the nutrition estimation branch while maintaining stable convergence. Compare with the previous classification-focused configuration, the revised loss balance resulted in positive R^2 values across all five nutrients. This finding demonstrates that the allocation of task-level loss weights can have a substantial effect on the representations learned by the shared backbone and on the final regression performance. Overall, both architectures exhibited stable regression learning, although they achieved this behavior using different task-loss balances. MobileNetV2 remained stable with greater classification emphasis, whereas ResNet50 benefited from stronger regression emphasis.

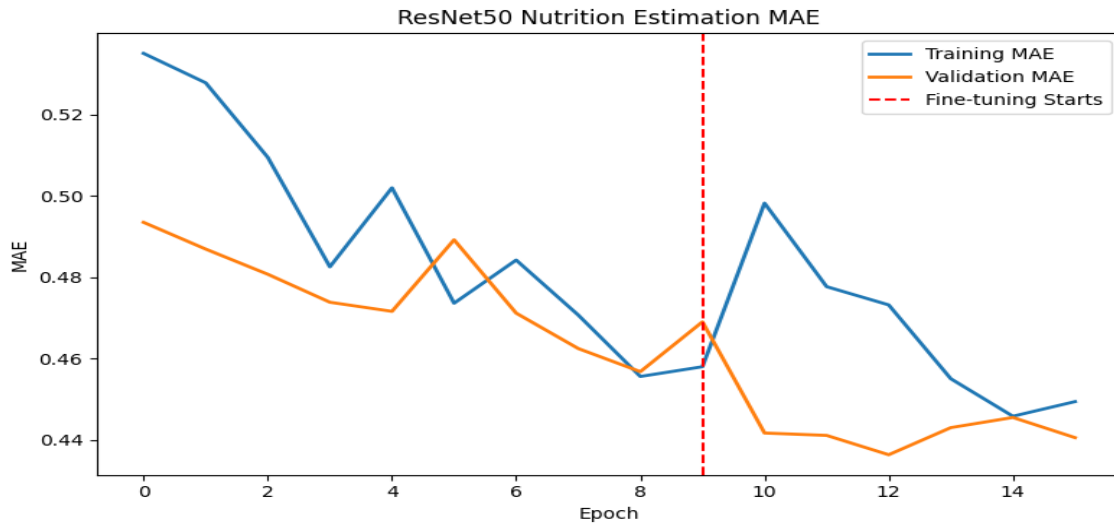


Figure 4.9: Nutrition Estimation MAE of ResNet50

4.3.3. Predicted versus Actual Nutritional Values

Predicted-versus-actual scatter plots provide a visual assessment of the regression performance by comparing the predicted nutritional values against their corresponding ground-truth values. In an ideal regression model, all observations would lie on the diagonal reference line, indicating perfect agreement between prediction and actual measurement. Therefore, the distance of each observation from the reference line reflects the magnitude of prediction error. Figures 4.10 and 4.11 illustrate the predicted-versus-actual relationships obtained by MobileNetV2 and ResNet50, respectively.

- **MobileNetV2**

Figure 4.10 presents the predicted-versus-actual relationships obtained using MobileNetV2. Sodium shows the strongest alignment with the ideal reference line, consistent with its R^2 value of 0.613. Carbohydrates also exhibit a visible positive relationship, although greater dispersion is observed for dishes with higher nutritional values. Calories and fats demonstrate moderate predictive capability, while protein shows weaker agreement and greater variability. Across the five targets, prediction errors generally increase for dishes with larger nutritional values. This pattern suggests that MobileNetV2 captures typical nutritional ranges more effectively than extreme values, which may be associated with uncommon serving sizes or visually ambiguous portion quantities.

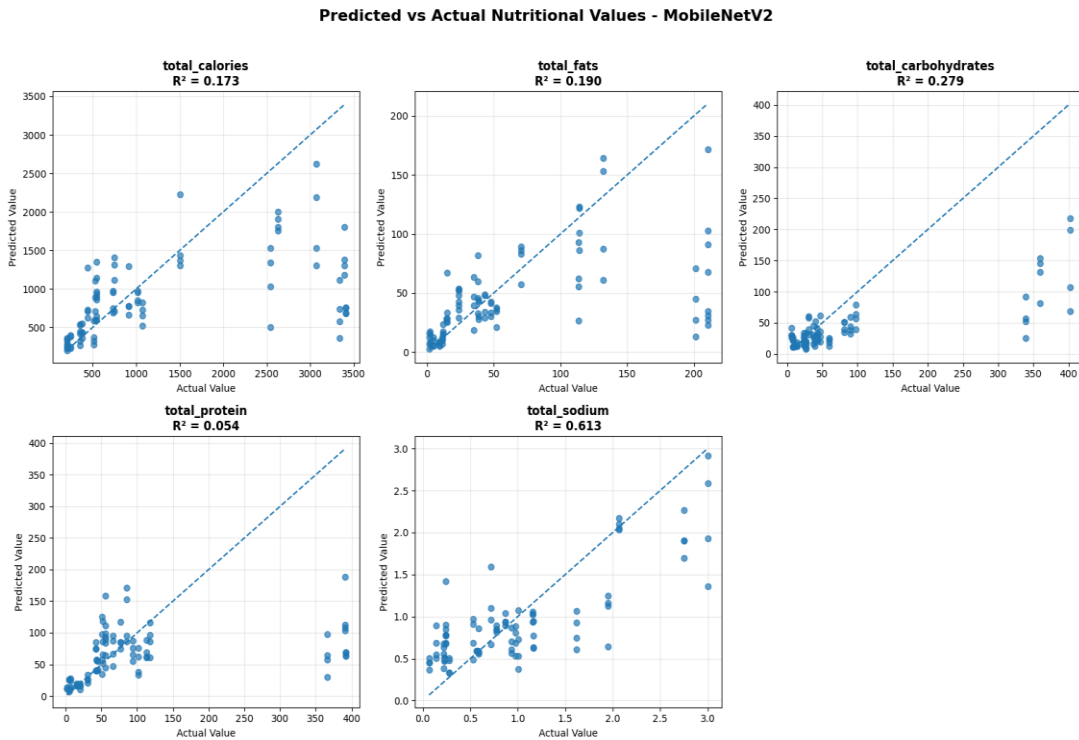


Figure 4.10: Predicted versus Actual Nutritional Values Using MobileNetV2

Among all nutritional targets, total sodium exhibited the strongest agreement between predicted and actual values, achieving the highest coefficient of determination ($R^2 = 0.613$). Most observations were distributed relatively close to the diagonal reference line, indicating that the model successfully learned the relationship between image features and sodium content. This result suggests that sodium is more closely associated with food category information, allowing the regression branch to benefit from the contextual features provided by the classification branch. The prediction performance for total carbohydrates was moderate, with an R^2 value of 0.279. Although a positive linear relationship between predicted and actual values can be observed, larger deviations from the reference line become apparent for dishes containing higher carbohydrate contents. These observations indicate that the model was capable of estimating carbohydrate content reasonably well for common serving sizes but encountered greater uncertainty when predicting dishes with substantially larger nutritional values.

For total calories and total fats, the regression performance was comparatively weaker, achieving R^2 values of 0.173 and 0.190, respectively. The scatter plots reveal greater dispersion around the diagonal line, particularly for dishes with high caloric and fat contents. In several

cases, the model underestimated dishes with extremely large nutritional values, suggesting that image appearance alone was insufficient to accurately infer the corresponding nutritional quantities. Among the five targets, total protein achieved the lowest prediction performance ($R^2 = 0.054$). The scatter distribution of observations indicates substantial variability between predicted and actual values, demonstrating that protein content could not be reliably estimated from RGB image features alone. One possible explanation is that protein content depends heavily on ingredient composition and serving quantity, which cannot be fully represented by two-dimensional visual information.

Overall, Figure 4.10 demonstrates that MobileNetV2 achieved the most reliable predictions for sodium and carbohydrates while showing reduced performance for calories, fats, and protein. These findings are consistent with the quantitative evaluation presented in Section 4.3.1 and indicate that nutrients strongly influenced by portion size remain considerably more challenging to estimate from a single RGB image.

- **ResNet50**

Figure 4.11 presents the predicted-versus-actual relationships obtained using ResNet50 under the revised regression-focused loss configuration. Sodium demonstrates the strongest agreement with the reference line and achieves the highest R^2 value in the study at 0.691. Carbohydrates also exhibit a clear positive relationship with an R^2 of 0.291, slightly exceeding the corresponding MobileNetV2 result. Calories achieve moderate predictive capability with an R^2 of 0.166, closely matching MobileNetV2. Protein improves to 0.072 and slightly exceeds the MobileNetV2 result, although substantial scatter remains. Fats achieve the lowest ResNet50 performance at 0.049, indicating only limited agreement between predicted and actual values.

The positive R^2 values across all five targets demonstrate that the increased regression task weight substantially improved the ability of ResNet50 to capture nutritional variation. Nevertheless, greater prediction dispersion remains visible for high-value samples, particularly for calories, fats, and protein, highlighting the continuing influence of portion ambiguity.

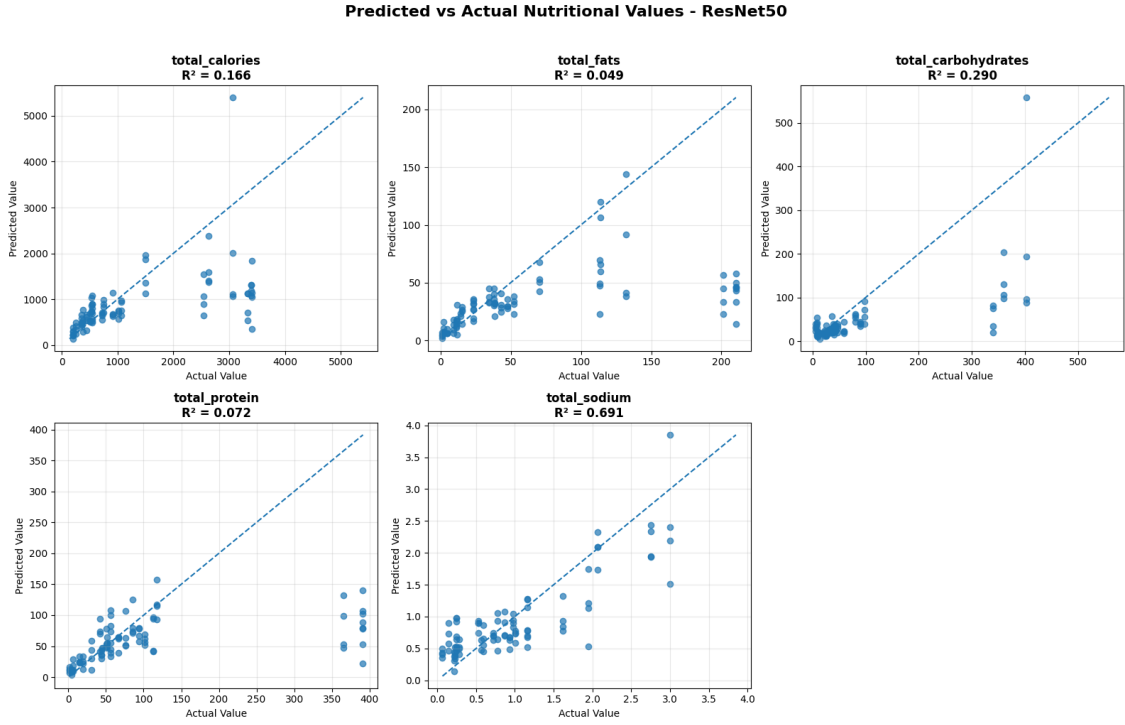


Figure 4.11: Predicted versus Actual Nutritional Values Using ResNet50

- **Comparative Discussion**

Comparing Figures 4.10 and 4.11 shows that neither architecture consistently dominates across all nutritional targets. MobileNetV2 demonstrates stronger agreement for calories and fats, whereas ResNet50 provides closer alignment for carbohydrates, protein, and especially sodium. The two models show nearly equivalent calorie prediction performance, with R^2 values of 0.173 and 0.166. These results indicate that backbone architecture and task-loss weighting jointly influence regression performance. The revised ResNet50 configuration demonstrates that its deeper visual representations can support both classification and nutrition estimation when sufficient optimization emphasis is assigned to the regression objective. However, the persistent dispersion observed for portion-dependent nutrients confirms that loss balancing cannot fully compensate for the absence of explicit serving-size and volume information.

4.3.4. Residual Analysis

Residual analysis was conducted to further evaluate the prediction behavior of the proposed regression models. Unlike the predicted-versus-actual plots presented in the previous section, residual plots provide insight into the distribution of prediction errors by illustrating the difference between the actual and predicted nutritional values. Ideally, residuals should be randomly distributed around the

zero references line without exhibiting systematic patterns. Such a distribution indicates that the regression model produces unbiased predictions across the range of nutritional values. Figures 4.12 and 4.13 present the residual plots obtained by MobileNetV2 and ResNet50, respectively.

- **MobileNetV2**

Figure 4.12 illustrates the residual distributions obtained by MobileNetV2 for the five nutritional targets. Overall, the residuals are centered around the zero referenced line for many testing samples, suggesting that the proposed regression model does not exhibit a consistent tendency to overestimate or underestimate nutritional values. Nevertheless, the magnitude of residuals varies considerably among the five nutrients.

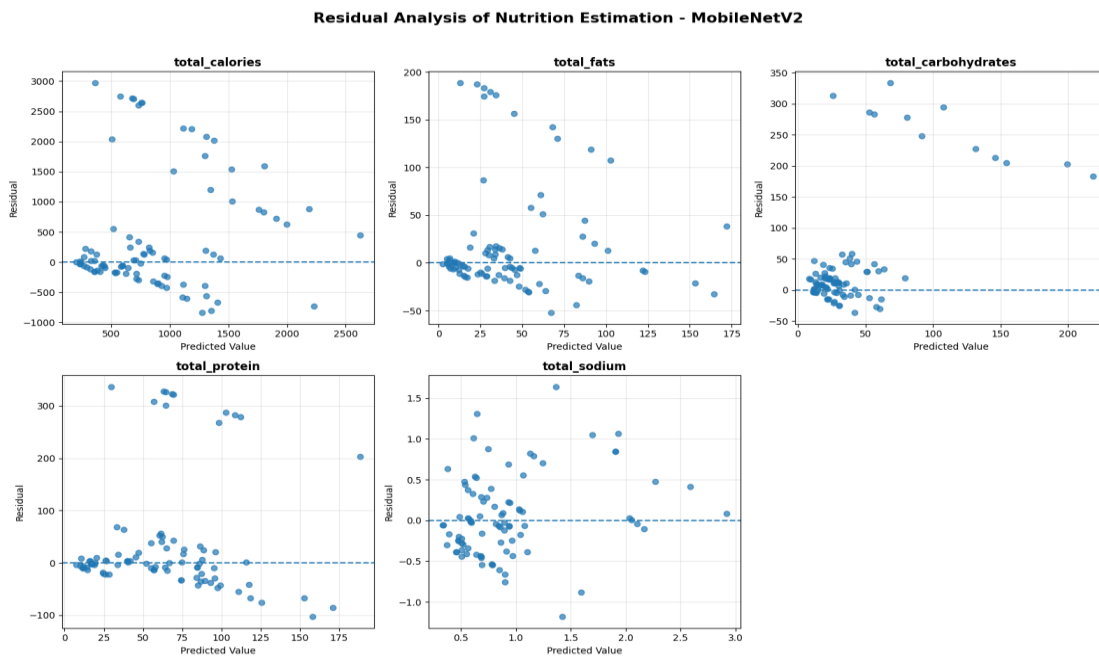


Figure 4.12: Residual Analysis of Nutrition Estimation Using MobileNetV2

For total sodium, the residuals are relatively concentrated around zero with a comparatively narrow spread. This behavior is consistent with the highest regression performance obtained in Section 4.3.1 ($R^2 = 0.613$), indicating that MobileNetV2 produced relatively stable predictions for sodium across different food images. Although several observations with larger positive and negative residuals remain, the overall dispersion is noticeably smaller than that of the other nutritional targets. The residual distributions for total carbohydrates and total fats exhibit moderate variability. Most observations remain close to the zero referenced line for dishes with relatively small predicted values; however, larger residuals become apparent as the predicted nutritional values increase. This trend suggests that the

regression model maintains reasonable prediction accuracy for common serving sizes but experiences increasing uncertainty when estimating dishes containing substantially larger nutritional quantities.

For total calories, the residual plot demonstrates the largest error dispersion among all nutritional targets. Several observations exhibit extremely large positive residuals, indicating that the model substantially underestimated the calorie content of certain dishes. This behavior reflects the inherent difficulty of estimating caloric values solely from RGB appearance, as calorie content is strongly influenced by portion size, food volume, and preparation methods that cannot be accurately inferred from a single two-dimensional image.

Similarly, total protein presents considerable residual variability despite relatively small residuals for many low-protein dishes. Several observations exhibit large positive residuals corresponding to protein-rich meals, suggesting that MobileNetV2 was less effective in estimating dishes containing high protein content. This finding is consistent with the relatively low coefficient of determination reported previously. Overall, the residual analysis demonstrates that MobileNetV2 provides relatively stable predictions for sodium and moderate performance for carbohydrates, whereas prediction errors increase substantially for calories, fats, and protein as nutritional values become larger.

- **ResNet50**

Figure 4.13 presents the residual distributions obtained using the revised ResNet50 model. Sodium exhibits the most compact residual pattern, consistent with its comparatively strong R^2 of 0.691 and low MAE of 0.35. Carbohydrates also show moderate concentration around the zero references line, although larger residuals remain for dishes with high carbohydrate values. The residuals for calories, fats, and protein are more dispersed, reflecting their lower coefficients of determination. In particular, several high-value samples exhibit large positive residuals, indicating underestimation of nutritionally dense or larger-portion dishes. Negative residuals also occur, showing that the model occasionally overestimates the nutritional content of visually similar samples.

Despite this variability, the residual patterns are no longer consistent with performance below the mean baseline because all five targets now achieve positive R^2 values. The results instead indicate that ResNet50 captures meaningful predictive structure but remains less reliable for extreme portions and visually ambiguous nutritional quantities.

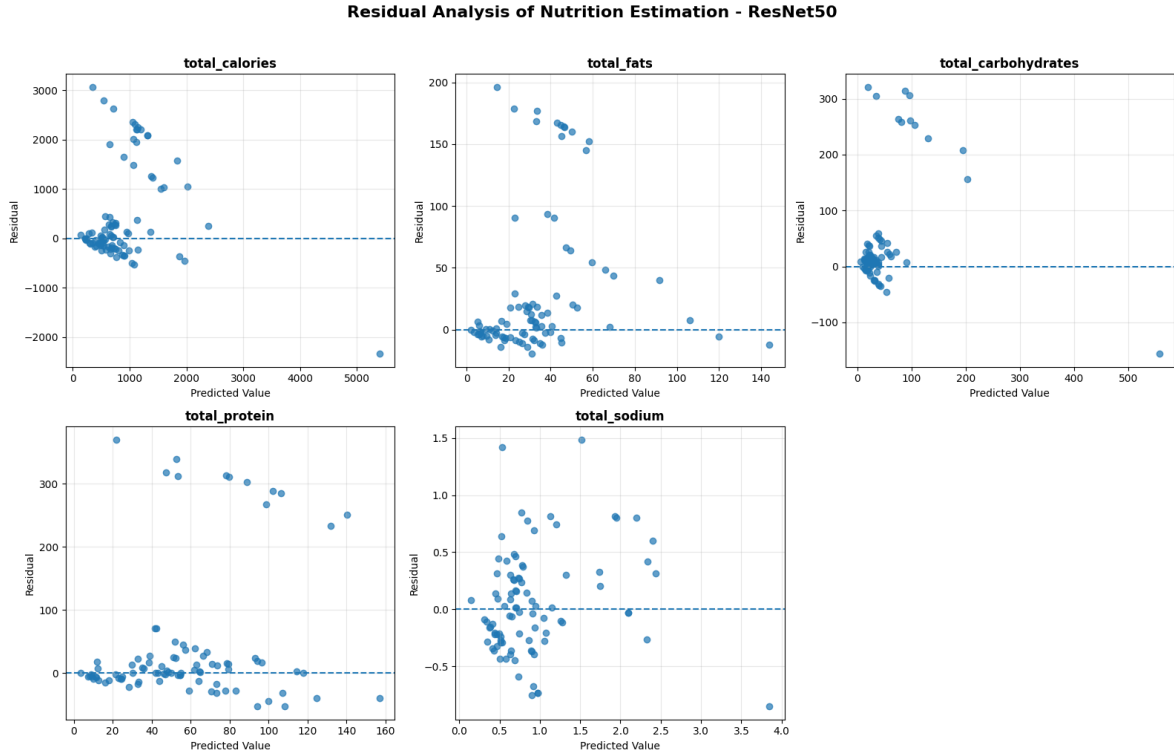


Figure 4.13: Residual Analysis of Nutrition Estimation Using ResNet50

- **Comparative Discussion**

Comparing Figures 4.12 and 4.13 demonstrates that MobileNetV2 generally produced more compact residual distributions than ResNet50, particularly for sodium, carbohydrates, and calories. Although both models experienced increasing prediction errors for dishes with large nutritional values, the residuals generated by MobileNetV2 were generally less dispersed, indicating more consistent regression behavior across the testing dataset. A common observation for both models is that prediction errors increased as the nutritional values became larger. This phenomenon is particularly evident for calories and fats, where several samples exhibit large positive residuals, indicating substantial underestimation of highly nutritious dishes. These findings suggest that the primary limitation of the proposed framework is not the optimization strategy but rather the intrinsic difficulty of inferring portion-dependent nutritional quantities from a single RGB image. Without explicit information regarding food volume, serving size, or depth, visually similar dishes may contain substantially different nutritional values despite appearing nearly identical.

In contrast, sodium consistently exhibited the smallest residual dispersion for both backbone networks, confirming that sodium content was more predictable from the available

visual information. This observation is consistent with the predicted-versus-actual analysis presented in Section 4.3.3 and the regression metrics reported in Section 4.3.1, where sodium achieved the highest coefficient of determination for both MobileNetV2 and ResNet50.

Overall, the residual analysis complements the quantitative regression evaluation by demonstrating that MobileNetV2 provides more stable and reliable nutritional predictions than ResNet50 within the proposed multi-task learning framework. Furthermore, the results reinforce the conclusion that accurate nutrition estimation from two-dimensional food images remains fundamentally constrained by the absence of explicit three-dimensional information describing food portion size and volume.

4.3.5. Overall Discussion

Overall, the regression results demonstrate that both MobileNetV2 and ResNet50 are capable of estimating multiple nutritional values directly from single 2D food images within the proposed multi-task learning framework. The regression-oriented task-loss configuration substantially strengthened ResNet50’s nutrition estimation performance, enabling it to achieve positive R^2 values across all five nutritional targets. Both architectures achieved positive R^2 values across all five nutritional targets, indicating that meaningful relationships between visual food features and nutritional information were successfully learned. Although neither backbone consistently outperformed the other across all nutritional targets, each architecture exhibited distinct strengths. MobileNetV2 achieved higher prediction accuracy for calories and fats, whereas ResNet50 produced superior results for carbohydrates, protein, and sodium. These findings suggest that the optimal balance between classification and regression objectives depends on the underlying backbone architecture. Assigning a greater regression contribution ($w_1 = 0.3, w_2 = 0.7$) enabled ResNet50 to better exploit its deeper residual feature representations for nutrition estimation, while MobileNetV2 maintained competitive performance using a classification-oriented weighting configuration.

Despite these improvements, the prediction accuracy remained inconsistent across different nutrients. Sodium achieved the highest R^2 values for both architectures, whereas calories, fats, and protein remained more challenging. This behavior reflects the inherent limitations of estimating nutritional information solely from RGB images, where portion size, food thickness, ingredient distribution, and occluded components cannot be accurately inferred. Consequently, prediction errors become more pronounced for nutrients that depend heavily on the physical quantity of food rather than its visual appearance. Overall, these

findings demonstrate that the proposed multi-task learning framework successfully balances food recognition and nutrition estimation while highlighting the importance of architecture-specific task-loss weighting. The results further indicate that improving optimization strategies can substantially enhance nutrition estimation performance without sacrificing stable model convergence, providing a promising direction for future vision-based dietary assessment systems.

4.4. Discussion

The experimental results presented in the previous sections demonstrate the effectiveness of the proposed multi-task learning framework for simultaneous food recognition and nutrition estimation from two-dimensional food images. While Sections 4.2 and 4.3 evaluated the classification and regression tasks independently, this section integrates these findings to provide a comprehensive comparison between the two backbone architectures, relates the observed results to previous studies, and discusses the practical implications, limitations, and future research opportunities arising from this work.

4.4.1. Comparison between MobileNetV2 and ResNet50

The comparative evaluation demonstrates that both MobileNetV2 and ResNet50 are effective backbone architectures for the proposed multi-task learning framework, although their strengths differ depending on the learning objective. For food recognition, ResNet50 achieved slightly better overall classification performance, obtaining a higher Micro Recall (0.5273), Micro F1-score (0.4979), and Macro F1-score (0.3905), indicating a stronger ability to identify multiple food components within complex dishes. In contrast, MobileNetV2 achieved slightly higher Micro Precision (0.5034), Hamming Accuracy (0.9132), and Micro AUC (0.8938), demonstrating more conservative and accurate predictions with fewer false positive classifications. These results indicate that both architectures provide comparable classification capability, with ResNet50 favoring recall and balanced recognition performance, while MobileNetV2 emphasizes prediction precision.

The comparison of nutrition estimation further highlights the complementary strengths of the two backbone architectures. MobileNetV2 achieved higher prediction accuracy for calories ($R^2 = 0.173$) and fats ($R^2 = 0.190$), whereas ResNet50 produced superior results for carbohydrates ($R^2 = 0.291$), protein ($R^2 = 0.072$), and sodium ($R^2 = 0.691$). Although the average regression performance of both architectures was comparable, these results suggest

that deeper residual feature representations allow ResNet50 to capture more complex relationships for certain nutritional targets when sufficient emphasis is placed on the regression task. Conversely, the lightweight architecture of MobileNetV2 remained highly competitive for nutrients that are more strongly associated with overall visual appearance.

One important observation from this study is that the optimal task-loss weighting differs across backbone architectures. MobileNetV2 achieved its best overall performance using a classification-oriented weighting configuration ($w_1 = 0.7, w_2 = 0.3$), whereas ResNet50 demonstrated substantially improved regression performance when the regression contribution was increased ($w_1 = 0.3, w_2 = 0.7$). This finding indicates that applying identical optimization settings to different backbone networks may not fully exploit their respective representational capabilities. Instead, architecture-specific task balancing can provide a more effective trade-off between food recognition and nutrition estimation.

From a practical perspective, MobileNetV2 remains an attractive choice for deployment on mobile and resource-constrained devices because of its lightweight architecture, lower computational complexity, and competitive prediction performance. In contrast, ResNet50 is more suitable for applications where maximizing prediction accuracy for multiple nutritional components is prioritized over computational efficiency. Overall, the results demonstrate that neither backbone architecture universally outperforms the other; instead, each exhibits complementary strengths that make it suitable for different application scenarios. framework.

4.4.2. Comparison with Previous Studies

The experimental results obtained in this research are generally consistent with previous studies demonstrating the effectiveness of deep convolutional neural networks for food image analysis while also highlighting the advantages of multi-task learning for simultaneous food recognition and nutrition estimation. Earlier food recognition studies primarily focused on image classification, whereas nutrition estimation was often treated as a separate regression problem or required additional information such as depth images, segmentation masks, or manually estimated portion sizes. In contrast, the proposed framework performs both tasks simultaneously using only a single RGB food image, providing a more practical solution for real-world dietary assessment.

Compared with previous image-based nutrition estimation approaches, the proposed framework demonstrates competitive performance despite relying exclusively on monocular

2D images. The positive R^2 values achieved across all five nutritional targets indicate that meaningful nutritional information can be learned directly from visual food features without requiring additional sensing modalities. In particular, the improved performance of ResNet50 following architecture-specific task-loss weighting demonstrates that optimization strategy plays an equally important role as backbone selection in multi-task learning. This finding extends previous work by showing that appropriate balancing between classification and regression objectives can substantially improve nutrition estimation performance. Nevertheless, the regression results also confirm observations reported in previous literature that predicting nutritional values solely from RGB food images remains inherently challenging. Nutrients such as calories, fats, and protein depend heavily on portion size, food thickness, ingredient proportions, and hidden food components that cannot be fully inferred from a single two-dimensional image. Consequently, prediction accuracy remains lower than food recognition performance despite stable optimization during training. Similar limitations have been discussed in previous studies, which suggest incorporating depth information, three-dimensional reconstruction, or volume estimation techniques to improve nutrient prediction accuracy.

Overall, the proposed multi-task learning framework contributes to the growing body of research on automated dietary assessment by demonstrating that architecture-specific task-loss weighting can significantly enhance nutrition estimation while maintaining competitive food recognition performance. These findings provide practical guidance for future research on designing balanced multi-task optimization strategies for vision-based nutrition assessment systems and support the continued development of accessible dietary monitoring technologies using ordinary RGB images.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

This research proposed an end-to-end multi-task deep learning framework for simultaneous food recognition and nutrition estimation from single 2D images of Asian food dishes. Using the NutritionVerse-Real dataset, the framework integrates a shared convolutional neural network backbone with separate classification and regression branches, enabling both tasks to be optimized jointly within a unified architecture. Two widely used convolutional neural network backbones, MobileNetV2 and ResNet50, were evaluated to investigate their effectiveness in supporting multi-task learning for automated dietary assessment. For food recognition, both backbone architectures achieved competitive classification performance. ResNet50 demonstrated stronger balanced recognition capability by achieving higher Micro Recall, Micro F1-score, and Macro F1-score, while MobileNetV2 achieved slightly higher Micro Precision, Hamming Accuracy, and Micro AUC. These results indicate that both architectures effectively learned discriminative visual representations for multi-label food recognition, although they exhibited different prediction characteristics.

For nutrition estimation, the proposed framework successfully learned meaningful relationships between food appearance and nutritional information. Both MobileNetV2 and ResNet50 achieved positive R^2 values across all five nutritional targets, demonstrating the feasibility of estimating nutritional values directly from single RGB food images. MobileNetV2 produced more accurate predictions for calories and fats, whereas ResNet50 achieved better performance for carbohydrates, protein, and sodium. The findings further demonstrate that architecture-specific task-loss weighting plays an important role in multi-task optimization. MobileNetV2 achieved its best performance using a classification-oriented task weighting ($w_1 = 0.7, w_2 = 0.3$), while ResNet50 benefited from a regression-oriented weighting ($w_1 = 0.3, w_2 = 0.7$), substantially improving its nutrition estimation capability without compromising stable model convergence.

Overall, the proposed multi-task learning framework achieved the research objectives by demonstrating that simultaneous food recognition and nutrition estimation can be effectively performed using only single RGB food images as input. Although prediction accuracy remains constrained by the inherent limitations of monocular images, the proposed approach provides

a practical and computationally efficient foundation for automated dietary assessment systems and supports future applications in nutrition monitoring, healthcare, and intelligent dietary recommendation.

5.2. Research Contributions

This research makes several important contributions to the field of computer vision-based dietary assessment. First, it proposes a unified end-to-end multi-task deep learning framework that simultaneously performs food recognition and nutrition estimation using only a single RGB food image. Unlike many previous studies that treat these tasks independently or require additional sensing modalities, the proposed framework jointly optimizes both objectives within a shared feature learning architecture. Second, this study demonstrates the effectiveness of architecture-specific task-loss weighting in multi-task learning. Experimental results show that different backbone architectures require different balances between classification and regression objectives to achieve optimal performance. This finding provides practical guidance for designing more effective multi-task learning models beyond the food analysis domain. Third, the proposed framework provides a comprehensive comparative evaluation between MobileNetV2 and ResNet50 under identical experimental conditions. The results reveal that the two backbone architectures exhibit complementary strengths rather than a single universally superior model. MobileNetV2 offers higher computational efficiency and stronger prediction for calories and fats, whereas ResNet50 provides improved balanced classification performance and superior prediction for carbohydrates, protein, and sodium. Finally, the research demonstrates the practical feasibility of automated nutrition estimation from ordinary RGB food images without requiring depth sensors, volume estimation devices, or manual portion measurements, making the proposed framework suitable for future deployment in accessible dietary monitoring applications.

5.3. Research Limitations

Despite the encouraging results, several limitations remain. First, nutrition estimation was performed using only single two-dimensional RGB food images. Because monocular images do not contain explicit depth or volume information, accurately estimating portion size remains difficult, particularly for nutrients that depend strongly on the physical quantity of food. Second, the NutritionVerse-Real dataset contains only 42 food categories after data cleaning. Although these food categories represent common Asian dishes, they cannot fully capture the diversity of regional cuisines, cooking styles, ingredient combinations, and

presentation variations encountered in real-world environments. Third, although the proposed multi-task framework successfully balances food recognition and nutrition estimation, prediction accuracy varies across different nutritional targets. Calories, fats, and protein remain more challenging to estimate than sodium because their values are more strongly influenced by hidden ingredients and serving size, which cannot be directly inferred from a single RGB image. Finally, this research evaluates only two convolutional neural network backbone architectures. Recent vision transformer architectures and hybrid CNN-transformer models may provide stronger feature extraction capabilities for multi-task learning and warrant further investigation.

5.4. Future Research

Several directions can be explored to further improve the proposed framework. First, future research should investigate the integration of depth estimation, three-dimensional reconstruction, or monocular volume estimation techniques to improve portion size estimation and consequently increase nutrition prediction accuracy. Second, expanding the dataset to include a larger variety of Asian and Cambodian foods, together with more diverse serving sizes, lighting conditions, and viewpoints, would improve the robustness and generalization capability of the proposed model in practical applications. Third, future work may explore more advanced backbone architectures, including EfficientNet, ConvNeXt, Vision Transformers (ViT), and hybrid CNN-transformer models, to further improve feature extraction and multi-task learning performance. Fourth, adaptive multi-task optimization methods, such as dynamic loss weighting or uncertainty-based loss balancing, could be investigated to automatically determine the optimal balance between classification and regression tasks during training instead of relying on manually selected task weights. Finally, the proposed framework could be integrated into mobile health applications or intelligent dietary monitoring systems capable of providing real-time food recognition, nutrition estimation, and personalized dietary recommendations to support healthier eating habits.

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